

REVIEW ARTICLE

The *locus* of Action of CGRPergic Monoclonal Antibodies Against Migraine: Peripheral Over Central Mechanisms

Abimael González-Hernández¹, Bruno A. Marichal-Cancino², Enrique García-Boll¹ and Carlos M. Villalón^{3,*}

¹Instituto de Neurobiología, Universidad Nacional Autónoma de México, Blvd. Juriquilla 3001, 76230 Querétaro, México; ²Departamento de Fisiología y Farmacología, Universidad Autónoma de Aguascalientes, Ciudad Universitaria, 20131 Aguascalientes, México; ³Departamento de Farmacobiología, Cinvestav-Coapa, Czada. de los Tenorios 235, Col. Granjas-Coapa, Deleg. Tlalpan, 14330 Ciudad de México, México

ARTICLE HISTORY

Received: November 30, 2019
Revised: April 18, 2020
Accepted: May 05, 2020

DOI:
10.2174/1871527319666200618144637

Abstract: Migraine is a complex neurovascular disorder characterized by attacks of moderate to severe unilateral headache, accompanied by photophobia among other neurological signs. Although an arsenal of antimigraine agents is currently available in the market, not all patients respond to them. As Calcitonin Gene-Related Peptide (CGRP) plays a key role in the pathophysiology of migraine, CGRP receptor antagonists (gepants) have been developed. Unfortunately, further pharmaceutical development (for olcegepant and telcagepant) was interrupted due to pharmacokinetic issues observed during the Randomized Clinical Trials (RCT). On this basis, the use of monoclonal antibodies (mAbs; immunoglobulins) against CGRP or its receptor has recently emerged as a novel pharmacotherapy to treat migraines. RCT showed that these mAbs are effective against migraines producing fewer adverse events. Presently, the U.S. Food and Drug Administration approved four mAbs, namely: (i) erenumab; (ii) fremanezumab; (iii) galcanezumab; and (iv) eptinezumab. In general, specific antimigraine compounds exert their action in the trigeminovascular system, but the *locus* of action (peripheral vs. central) of the mAbs remains elusive. Since these mAbs have a molecular weight of ~150 kDa, some studies rule out the relevance of their central actions as they seem unlikely to cross the Blood-Brain Barrier (BBB). Considering the therapeutic relevance of this new class of antimigraine compounds, the present review has attempted to summarize and discuss the current evidence on the probable sites of action of these mAbs.

Keywords: Antibody, antimigraine, CGRP, gepants, headache, pain.

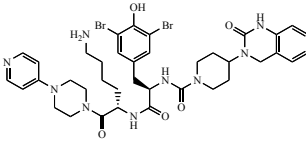
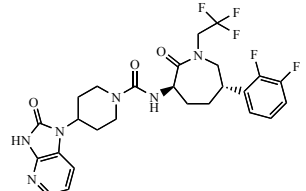
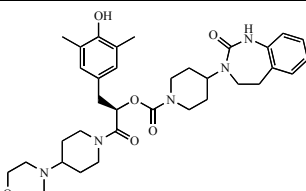
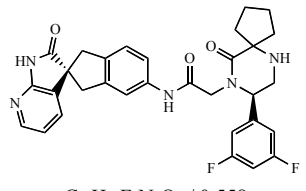
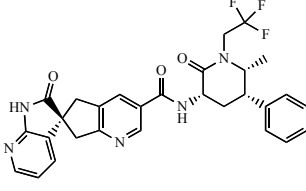
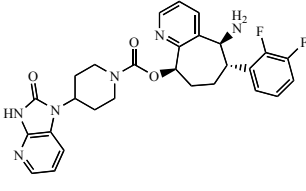
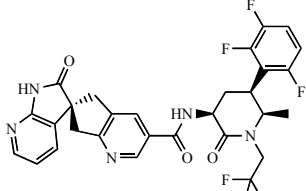
1. GENERAL INTRODUCTION

Migraine is a severe and disabling chronic brain condition [1, 2] with a strong neurovascular component [3-5]. It is currently considered one of the most prevalent medical conditions and the second most disabling neurological disorder in the world [6, 7]. This disorder is characterized by unilateral throbbing headaches and a wide range of neurological symptoms such as hypersensitivity to light and sound, nausea and a variety of autonomic, cognitive, emotional and motor disturbances [2]. Although several drugs to treat migraines are now available (prophylactic and acute), their utility is often limited by side effects such as nausea, somnolence and/or dizziness [8]. At present, Calcitonin Gene-Related Peptide (CGRP) and its receptor seem

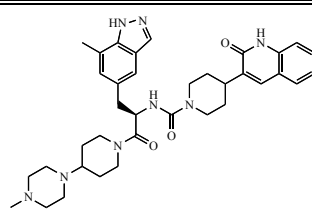
to be important therapeutic targets for the development of new pharmacotherapies against this disorder [9]. In the last thirty years, CGRP (a potent vasodilator released from sensory nerves) has been described as a key molecule involved in the pathophysiology of migraine [4]. In addition to cranial vasoconstriction, one of the most important mechanisms associated with the therapeutic effects of acute antimigraine agents like the triptans (e.g., sumatriptan) is their capability to interfere with the CGRPergic transmission at peripheral and central levels [10]. Following the idea of specificity, further development of agents targeting CGRPergic transmission has emerged (Table 1). In the last twenty years, antagonists (*i.e.*, gepants) of the CGRP receptor have been developed, but some of these drugs (*i.e.*, olcegepant and telcagepant) have been withdrawn from the clinical use due to pharmacokinetic/hepatotoxicity concerns [9]. More recently, the development of monoclonal antibodies (mAbs) as a new strategy to interfere with the CGRPergic transmission by blocking the function of CGRP or the CGRP

*Address correspondence to this author at the Departamento de Farmacobiología, Cinvestav-Coapa, Czada. de los Tenorios 235, Col. Granjas-Coapa, Deleg. Tlalpan, 14330 Ciudad de México, México; Tel: (52)-(55)-5483-2854; Fax: (52)-(55)-5483-2863; E-mail: cvillalon@cinvestav.mx

Table 1. Small non-peptide molecules (gepants) and monoclonal antibodies (mAbs) against the CGRPergic transmission.

	Compound (Company)	Other Names	Status	Route of Admin- istration/ Indication	Formula / m.w. (kDa)
CGRP receptor antagonists; pK _i > 9.0 (small molecules, m.w. < 0.9 kDa)	OLCEGEPANT (Boehringer Ingelheim)	BIBN4096BS	Completed phase II (discontinued, low bioavailability)	Intravenous/acute treatment	 C ₃₈ H ₄₇ Br ₂ N ₉ O ₅ / 0.870
	TELCEGEPANT (Merck & Co.)	MK-0974	Completed phase III (discontinued, hepa- totoxicity concerns)	Oral/acute treatment	 C ₂₆ H ₂₇ F ₅ N ₆ O ₃ / 0.567
	BI 44370 TA (Boehringer Ingelheim)	-	Completed phase II (discontinued, hepa- totoxicity concerns)		 C ₃₅ H ₄₇ N ₅ O ₆ / 0.633
	MK-3207	-	Completed phase II (discontinued, hepa- totoxicity concerns)		 C ₃₁ H ₂₉ F ₂ N ₅ O ₃ / 0.558
	UBROGEPANT (Allergan)	MK-1602	FDA (2019) APPROVED		 C ₂₉ H ₂₆ F ₃ N ₅ O ₃ / 0.549
	RIMEGEPANT (Biohaven Pharmaceuticals)	BMS-927711 / BHV-3000	FDA (2020) APPROVED		 C ₂₈ H ₂₈ F ₂ N ₆ O ₃ / 0.535
	ATOGEPAANT (Allergan)	AGN-241689	Completed phase III (under FDA revision)	Oral/prophylactic treatment	 C ₂₉ H ₂₃ F ₆ N ₅ O ₃ / 0.603

(Table 1) Contd....

	Compound (Company)	Other Names	Status	Route of Admin- istration/ Indication	Formula / m.w. (kDa)
	VAZEPEGANT (Biohaven Pharmaceuticals)	BHV-3500	Under Phase II/III	Intranasal / acute treatment	 $C_{36}H_{46}N_8O_3$ / 0.639
mAbs blocking the CGRPergic transmission; pK_i > 9.0 (protein structure. m.w. ~ 150 kDa)	ERENUMAB (Novartis AG/Amgen Inc.)	AMG-334	FDA (2018)/EMA (2018) APPROVED	Subcutaneous (monthly) / preemp- tive treatment	$C_{6472}H_{9964}N_{1728}O_{2018}S_{50}$ / ~150
	GALCANEZUMAB (Eli Lilly & Co.)	LY2951742	FDA (2018) APPROVED		$C_{6392}H_{9854}N_{1686}O_{2018}S_{46}$ / ~144
	FREMANEZUMAB (Teva Pharmaceutical Industries Ltd.)	RN-307 / LBR-101 / TEV-48125	FDA (2018)/EMA (2019) APPROVED		$C_{6470}H_{9952}N_{1716}O_{2016}S_{46}$ / ~145
	EPTINEZUMAB (H. Lundbeck A/S)	ALD 403	FDA (2020) APPROVED	Intravenous (quarter- ly) / preemptive treatment	$C_{6352}H_{9838}N_{1694}O_{1992}S_{46}$ / ~143

receptor is highly promising; indeed, in the last two years, four mAbs have been approved by the U.S. Food and Drug Administration (FDA) for the preventive treatment of migraine (Table 2).

2. A BRIEF ACCOUNT ON MIGRAINE PATHOPHYSIOLOGY

Based on the clinical features of migraine, three distinct phases can be discerned: (i) a trigger, (ii) an aura, and (iii) a headache phase [9]. The characteristics of these phases are beyond the objective of the present review. However, the take-home message is that the migraine trigger involves abnormal neuronal processing (due to genetic factors, *i.e.*, a P/Q type Ca^{2+} channel dysfunction) in the brainstem and other central structures. Certainly, epigenetic modifications, particularly DNA methylation, could impair the function of some genes closely related with the transcription and synthesis of proteins involved in the pathophysiology of migraine [*e.g.*, genes related with the expression of nitric oxide synthase 1 (NOS1), α CGRP, or the Receptor Activity-Modifying Protein 1 (RAMP1)] [11]. In the case of the aura and headache phases, these events have been explained by the neurovascular theory [2, 4, 9].

Undeniably, the pathophysiology of migraine relies on a critical neurophysiological state where the function of the organism leads to a low plasma concentration of serotonin (5-hydroxytryptamine; 5-HT) and high plasma concentrations of CGRP [12]. These observations have led to the development and clinical current use of specific antimigraine compounds, such as: (i) triptans (5-HT_{1B/1D} receptor agonists; *e.g.*, sumatriptan); (ii) ditans (5-HT_{1F} receptor agonists; *e.g.*, lasmiditan); (iii) gepants (CGRP receptor antagonists; *e.g.*, ubrogepant); and (iv) mAbs (against CGRP or its receptor; *e.g.*, erenumab) [12].

Although the precise origin of the neuronal events underlying a migraine attack is unknown, it is accepted that, apart

from the altered excitability of supraspinal brain centers (*i.e.*, brainstem and diencephalic nuclei), migraine involves activation and sensitization of the trigeminovascular pathways [1, 2]. At the peripheral level, the sensitization of trigeminovascular pathways induces the release of CGRP from perivascular trigeminal sensory fibers and, consequently, vasodilatation of peripheral blood vessels, whereas, at central levels, facilitation of the nociceptive transmission could be induced by this neuropeptide [1, 4, 5, 10, 13].

2.1. The Relevance of the Trigemino-vascular System

The trigeminovascular system consists of nociceptive primary A δ - and C-fibers derived from the trigeminal ganglion; they innervate craniovascular structures. These primary afferent fibers converge upon the brainstem, mainly at the trigeminal nucleus *pars caudalis* (Sp5c), which, in conjunction with the upper cervical spinal cord (C₁-C₂) segments, comprises the Trigemino-cervical Complex (TCC), a key structure in cranial nociceptive processing. The TCC receives afferent inputs from meninges, extracranial and intracranial blood, facial skin and muscles, as well as from the greater occipital nerve [15, 16].

In humans, activation of the trigeminovascular system by electrical stimulation of the middle cerebral artery or middle meningeal artery elicits pain [17]. Accordingly, electrical stimulation (in cats) or thermocoagulation (in humans) of the trigeminal ganglion induces an increase in CGRP in the extracerebral circulation (measured at the external jugular vein) [14]. Moreover, in humans, immunoreactivity to CGRP was found at the TCC level [3]. In fact, upon stimulation of the trigeminal nerve, vasodilation of dural and pial blood vessels is induced, and a cascade of neuronal events results in perivascular changes (*i.e.*, vasodilatation) and sensitization of the trigeminovascular system [18, 19]. This highlights the close interaction between the neural innervation and the vasculature in the nociceptive processing of migraines.

Table 2. Current monoclonal antibodies (mAbs) clinically approved to treat migraines targeting the Calcitonin Gene-Related Peptide (CGRP) pathway (CLR-RAMP1 complex [CGRP receptor] or the human α - or β -CGRP). The data were taken in part from [116, 152, 153, 154, 155, 156]. EM, episodic migraine; CM, chronic migraine; eCH, episodic cluster headache; cCH, chronic cluster headache; s.c., subcutaneous; i.v., intravenous.

	Monoclonal Antibody			
Generic Name	ERENUMAB	GALCANESUMAB	FREMANESUMAB	EPTINEZUMAB
Brand Name	Aimovig®	Emgality®	Ajovy®	Vyepti™
Pharmacodynamic	Against CGRP receptor	Against CGRP	Against CGRP	Against CGRP
Indications	EM, CM	EM, CM, eCH, cCH	EM, CM, eCH, cCH	EM, CM
Interval between administration	4 weeks (s.c.)	4 weeks (s.c.)	4 or 12 weeks (s.c.)	12 weeks (i.v.)
Immunoglobulin type	IgG2 λ Fully human >90% homologous to <i>Homo sapiens</i> Produced in transgenic Xenomouse®	IgG4 Humanized >85% homologous to <i>Homo sapiens</i>	IgG2 κ Humanized >85% homologous to <i>Homo sapiens</i>	IgG1 κ Humanized >81% homologous to <i>Homo sapiens</i> Produced in yeast
T _{max} / Half-life (days)	3-14 / 21	7-14 / 28	3-20 / 40-48	4.8 hours / 31

2.2. The Neurovascular Theory of Migraine

The theories of migraine pathophysiology have changed over the years, from vascular to neuronal to neurovascular theories. In the beginning, the vascular theory of migraine pointed out the role of cranial vasodilatation in migraine [20]. Later, the participation of blood vessels was questioned, and the dogma of migraine centered in neurons led to a neuronal theory [21]. In our opinion, both neurons and cranial blood vessels are involved in migraine pathophysiology if we consider it a complete phenomenon, as one cannot categorically deny neuronal and vascular alterations [22, 23]. It is interesting to highlight that the receptors mediating vasoconstriction of meningeal blood vessels are the same as those inhibiting the pain transmission (*i.e.*, 5-HT_{1B/1D} receptors); thus, antimigraine drugs designed to produce cranial vasoconstriction also inhibit the nociceptive pathways [24].

Harold Wolff's seminal observations in the 1940s led to the postulation that vasodilation of extracranial and intracranial blood vessels produces mechanical activation of perivascular nerve fibers, resulting in headache pain [25]. In agreement with this view, antimigraine drugs such as the triptans (5-HT_{1B/1D} receptor agonists) produce cranial vasoconstriction [26] and, consequently, reduce the activation of perivascular nerve fibers. However, at TCC, 5-HT_{1B/1D/1F} receptors exist and, as demonstrated by Kaube *et al.* [27], under disruption of the Blood-Brain Barrier (BBB), *i.v.* sumatriptan (a non-liposoluble 5-HT_{1B/1D} receptor agonist) inhibited the nociceptive firing of TCC, pointing out the relevance of central mechanisms in the action of antimigraine compounds. Certainly, the idea that the BBB is disrupted during a migraine attack has been questioned, as discussed below (4.2.2. *The blood-brain barrier in migraine pharmacotherapy*). In any case, during migraine attacks, intracranial blood vessels show only a slight dilatation that is not affected by the triptans and, consequently, a hot debate exists on the relevance of peripheral vascular structures in migraine

pathophysiology (*i.e.*, central *versus* peripheral mechanisms) [21, 28-37].

During the last years, migraine is more commonly described in terms of the neurovascular theory, in which the peripheral sensitization of trigeminal nociceptive afferents that innervate dural blood vessels plays a key role in the central sensitization of the trigeminovascular system, favoring extracranial hypersensitivity symptoms in migraine (*i.e.*, allodynia to periorbital and cutaneous facial areas) [13, 38, 39]. Moreover, sustained activation of dural meningeal nociceptive neurons causes sequential activation and sensitization of first-, second- (at the level of the TCC) and third-order (trigeminohypothalamic) neurons [40], as well as ascending activation of other diencephalic structures [1].

2.2.1. The Role of 5-Hydroxytryptamine (5-HT)

During a migraine attack, high levels of 5-hydroxyindole acetic acid (a 5-HT metabolite) are excreted, and monoamine depletors induce migraine attacks that are aborted by intravenous infusions of 5-HT; these findings led to suppose that 5-HT receptors may be involved in migraine pathophysiology [12, 41]. 5-HT can bind to seven classes of receptors (5-HT₁ - 5-HT₇), but at craniovascular structures, 5-HT predominantly induces vasoconstriction of intra- and extracranial arteries by the activation of 5-HT_{1B/1D} receptors [12, 41, 42]. These findings, in addition to boosting the development of compounds with selectivity for 5-HT_{1B/1D} receptors such as the triptans (*e.g.*, sumatriptan), were also crucial to understand some aspects of migraine pathophysiology. Apart from the direct vasoconstrictor effect induced by the triptans, these compounds inhibit the CGRPergic vasodilation at the peripheral level (by inhibition of CGRP release from capsaicin sensory fibers), and the CGRP release at TCC (inhibiting the nociceptive neuronal firing) [52]. Unfortunately, in patients with cardiovascular diseases, the use of triptans is contraindicated, since these compounds moder-

ately constrict the coronary artery. To evade this inconvenience, selective 5-HT_{1F} receptor agonists (*i.e.*, ditans) have been proposed [43, 44]. In fact, at TCC level, serotonergic mechanisms mediated by 5-HT_{1B/1D} receptor activation inhibited nociceptive transmission [12, 28, 45], and further evidence for an exclusive neuronal mechanism that may be lacking vascular effects [46] is supported by the selective 5-HT_{1F} receptor agonists [LY334370 and LY57314 (lasmiditan)] [47-52].

2.2.2. The Key Role of Calcitonin Gene-Related Peptide (CGRP)

CGRP is a major player in migraine pathophysiology [9]. This neuropeptide is composed of 37 aminoacids, and its receptor is found in the TCC, in the spinal trigeminal tract and in the trigeminal ganglion, among other supraspinal structures (*e.g.*, periaqueductal gray matter, hypothalamus) relevant to migraine pathophysiology. CGRP was primarily described as a potent vasodilator, stored in dense-core vesicles within nociceptive nerve terminals and, upon its release, binds to its receptor which consists of three proteins, namely: (i) the Calcitonin Receptor-Like Receptor (CLR), (ii) the Receptor Activity Modifying Protein 1 (RAMP1) and (iii) the Receptor Component Protein (RCP). As mentioned above, this peptide induces vasodilation and sensitization of nociceptive pathways [23, 53, 54].

Experimental activation of the trigeminal ganglia in animals and humans results in the release of CGRP [12], which can be inhibited by triptans not only at Sp5c, but also in the dura mater and trigeminal ganglion [55]. Indeed, elevated CGRP plasma levels during migraine attacks can be normalized after sumatriptan [56]. Certainly, as recent migraine pharmacotherapies against the CGRPergic transmission are effective in the preventive treatment of migraine [57], it may be assumed that CGRP is involved in the triggering as well as in the maintenance of a migraine attack and its associated symptomatology. As a result, the blockade of CGRPergic mechanisms in the trigeminovascular system has resulted in an antimigraine action, and mAbs against CGRP or its receptor have been developed as a novel preventive treatment that represents an innovative biological approach for the migraine treatment [57-60].

3. CURRENT SPECIFIC MIGRAINE PHARMACOTHERAPY

3.1. Generalities

The current specific antimigraine pharmacotherapy is related to the activation of 5-HT_{1B/1D/1F} receptors and inhibition of CGRPergic transmission [13]. However, migraine is a complex pain entity, and unspecific pharmacotherapies have also been used. For example, patients with mild-to-moderate migraine attacks demand non-steroidal anti-inflammatory drugs (NSAIDs) [61]. From the economic point of view, these drugs that inhibit the activity of cyclooxygenase 1 and 2 are cheap and, consequently, are chosen as a first drug-class to treat migraines. In addition, NSAIDs such as ibuprofen have comparable relief effects to more specific antimigraine options [61]. For more intense migraine pain, the first specific pharmacotherapy for treating migraine came from rye ergot alkaloids, which produce (among other ef-

fects) potent vasoconstriction [62-64]. Ergotamine was then modified to dihydroergotamine to enhance the pharmacokinetics and to reduce the side effects [65-67]. As previously pointed out, one of the first mechanisms proposed for the antimigraine drugs was cranial vasoconstriction, but we currently know that ergots also inhibit the release of vasoactive neuropeptides and block the peripheral nociceptive integration at TCC by the activation of an array of monoaminergic receptors (serotonergic, noradrenergic and dopaminergic receptors) [64, 68, 69]. On the other hand, although there is an array of drugs that can stop or prevent migraine attacks, some patients are refractory to the current pharmacotherapy.

In view that some authors have proposed that migraine is a neuroinflammatory disorder, they suggest that migraine could be treated prophylactically by agents with anti-inflammatory action (indeed, NSAIDs are effective in treating migraine of mild to moderate severity). Particularly, considering that the Tumor Necrosis Factor- α (TNF- α) induces hyperexcitability of nociceptors, treatment with ω -3 fatty acids and curcumin (which acts by suppressing the TNF- α expression) reduces the frequency of migraine attacks [70-72]. Nonetheless, this kind of alternative treatment could be considered unspecific due to the pleiotropic functions of TNF- α . In any case, the current clinical pharmacological evidence supports the use of drugs interacting with 5-HT_{1B/1D/1F} receptors or interfering with the CGRPergic transmission as specific antimigraine agents.

3.2. An Outline of Drugs Designed to Stimulate 5-HT_{1B/1D/1F} Receptors

After the ergots' era, more selective antimigraine drugs were developed, such as the triptans (historical perspective [12, 41, 73]). Triptans are selective 5-HT_{1B/1D} receptor agonists that produce three main effects: (i) selective cranial vasoconstriction by direct vascular stimulation of 5-HT_{1B} receptors; (ii) inhibition of CGRP release (and consequently inhibition of vasodilation) at craniovascular level by the activation of perivascular 5-HT_{1B/1D} receptors; and (iii) inhibition of the nociceptive transmission in the trigeminovascular system [44]. These drugs were successfully incorporated into antimigraine therapeutics with good clinical results [74-76].

In general, sumatriptan, an effective and relatively safe triptan, has become a landmark in the treatment of migraine attacks due to its fast and good pain relief effect [77]. Nevertheless, some peculiarities, such as its low oral bioavailability and short half-life, encouraged the development of 5-HT_{1B/1D} receptor agonists with a better pharmacokinetic profile, the so-called "second generation" triptans (*e.g.*, eletriptan, frovatriptan, naratriptan, rizatriptan and almotriptan) [42, 78]. Unfortunately, 30 - 40% of migraine subjects are considered triptan-non responders [79], and even when triptans have been shown to be effective, their cardiovascular side effects attributable to vascular 5-HT_{1B/1D} receptor activation limit their use [43, 74]. However, since the triptans also have an affinity for the 5-HT_{1F} receptor subtype (a receptor found in the trigeminal ganglia and Sp5c), a new kind of drug (ditans) with preferential selectivity for the 5-HT_{1F} receptor has been developed. Indeed, lasmiditan failed to produce vasoconstriction [44], and was recently (October 2019) FDA-approved for migraine treatment. Notwithstand-

ing, in comparison with other triptans, the Central Nervous System (CNS) adverse events elicited by lasmiditan seem to be high, probably due to its liposolubility in conjunction with the fact that 5-HT_{1F} receptors are expressed in several brain areas [43, 44]. These data point out that further clinical monitoring of CNS events will be relevant.

3.3. Antagonists Against CGRP Receptors (Gepants)

Once a significant role for CGRP in migraine was spotted [14], and in view of the cardiovascular unspecific effects induced by the triptans that limited their use [77], a logical strategy for achieving a selective antimigraine therapy consisted in the synthesis of selective CGRP receptor antagonists (*i.e.*, gepants) (Fig. 1). These drugs are small molecule (m.w. < 0.9 kDa) non-peptide antagonists of the CGRP receptor that inhibit CGRP-induced vasodilation without vasoconstriction. Indeed, under healthy conditions, CGRP does not exert an appreciable effect on the vascular tone [43]. Hence, the current evidence on the gepants' cardiovascular safety suggests that acute blockade of the CGRPergic system in cardiovascular uncompromised subjects does not have a substantial impact [53]. However, as discussed elsewhere [53], CGRP plays a protective role against hypertension, myocardial infarction and heart failure. Indeed, in a recent study using a mouse surrogate model of stroke, olcegepant and rimegepant worsen the ischemic outcome induced by occlusion of the middle cerebral artery [80]. These data highlight the relevance of specific studies for evaluating the safety of new antimigraine compounds.

In this sense, eight gepants (Table 1) have been developed for the treatment of migraine [81]. In 2004, intravenous olcegepant showed efficacy against migraine [82], but it was not further developed as it cannot be orally administered. In 2008, an orally bioactive gepant (telcagepant) showed clinical efficacy in RCT [83]. Although these gepants were promising, pharmacokinetics (olcegepant) and hepatotoxicity (telcagepant) issues halted their further development [43]. Nevertheless, several orally bioavailable gepants (atogepant and vazegepant) are under RCT (phase II and/or III), showing promising results for the acute and preventive treatment of migraine [84-87]. Indeed, ubrogepant was approved for migraine treatment in December 2019 and rimegepant in February 2020 (Table 1).

3.4. Monoclonal Antibodies (mAbs) Against the CGRPergic Transmission

3.4.1. mAbs Against CGRP and the CGRP Receptor

Olcegepant and telcagepant highlight the relevance of the CGRPergic transmission in migraine headaches, but due to pharmacokinetics/hepatotoxic issues, their development was stopped. One interesting point in the use of these molecules as antimigraine agents is that telcagepant's antimigraine effect was associated with peripheral rather than central actions [88]; these findings pave the way for the idea that central actions are not required to alleviate migraine pain. The nature of the mAbs is such that they: (i) possess high molecular weight and, consequently, cannot cross the BBB (under

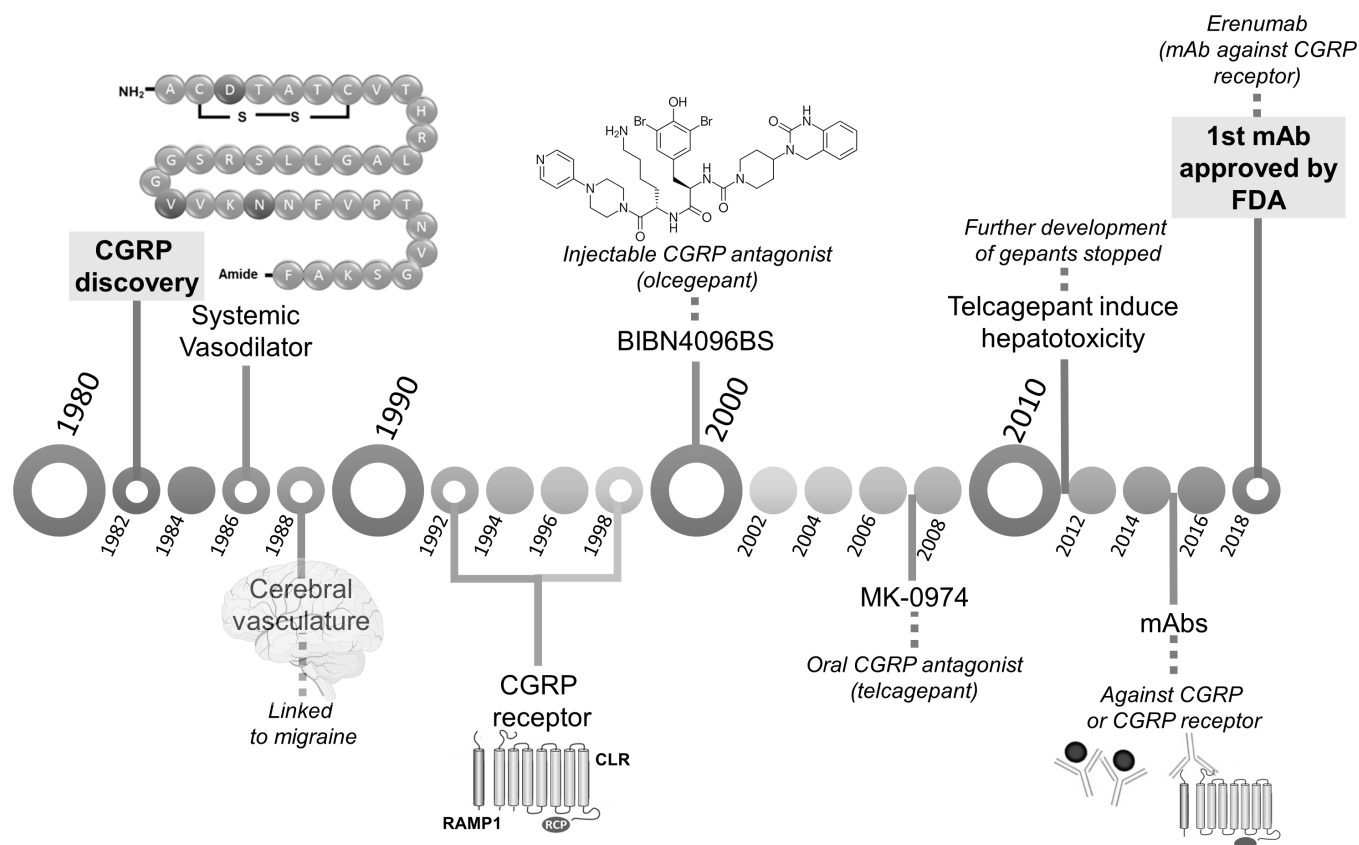


Fig. (1). A short timeline on CGRP discovery and migraine treatment. From its discovery to the novel mAbs approved for migraine treatment. (A higher resolution/colour version of this figure is available in the electronic copy of the article).

healthy conditions), and (ii) are “hepato-friendly” since they are not metabolized by this way. As a result, the use of mAbs emerged as a potential strategy to treat migraines. In general, these molecules act as “scavengers” of circulating CGRP molecules (nezumabs) or blockers of the CGRP receptor (numabs) [53]. Furthermore, the use of this type of immunoglobulins (IgG’s) as therapeutic agents seems to be well tolerated, and the most frequent adverse effects reported in the RCT for the mAbs against migraine were similar in proportion to those of the placebo treatment.

Currently, three humanized mAbs against CGRP (nezumabs) showed good results in RCT phase II and III studies [89-93], namely: (i) galcanezumab (LY2951742); (ii) fremanezumab (TEV-48125); and (iii) eptinezumab (ALD 403). These molecules are not capable of distinguishing between the rat and human α -CGRP or β -CGRP. In the case of galcanezumab, an increase of the upper respiratory tract infections has been reported [91, 94], probably because at pulmonary level, CGRPergic transmission is involved in airway inflammation [95, 96].

The human mAb against the CGRP receptor (numab) is erenumab (AMG-334) and was the first mAb approved (May 2020) for migraine treatment. Preventive treatment with erenumab in RCT phase II and III studies showed that this mAb is effective and well tolerated [98, 99]. In this case, and apart from the inflammation in the site of injection, no specific adverse events were reported [100]. Erenumab showed efficiency for decreasing the number of monthly migraine days in at least 50%, whereas tolerability and adverse events were not different from placebo [99-102].

3.4.2. An Overview of the Randomized Clinical Trial (RCT) of mAbs and Potential Adverse Events

In the last two years (May 2018 - February 2020), four mAbs against migraine were approved by the FDA (Table 2). Erenumab is considered a fully human mAb, whereas galcanezumab and fremanezumab are humanized mAbs. In general, mAbs have been shown to be well tolerated, and their beneficial effects last several weeks after administration [103]. Separation of effect from placebo for all mAbs occurs within the first week, and mAbs represent the most promising prophylactic therapeutics to deal with episodic and chronic migraine. These agents achieve a rapid effect over days to weeks. In addition, although CGRP is considered a relevant molecule in the cardiovascular system, efficacy and safety studies for these mAbs have not shown any relevant cardiovascular adverse events [52, 81, 104]. In fact, the RCT of mAbs exhibits a safety profile superior to all former migraine prophylactics [89, 91, 99, 101, 105, 106]. More recently, to evaluate the security of erenumab in patients with stable angina pectoris, an RCT phase II study has been launched and the results suggest that this antibody does not have any substantial adverse effects in this type of patients [107], but some concerns on the methodology used have been pointed out [108]. Admittedly, we do not have a clear idea of the risks associated with long-term blockade of CGRPergic signaling in the cardiovascular system, but the long-term mAbs treatment could favor an autoantibody production against mAbs, and this effect could be the most important adverse effect related with the use of these mAbs.

Experiments with a mirror-image CGRP binding oligonucleotide do not elicit any effect on the arterial diameter or blood flow [109]. Moreover, *in vitro* experiments show that although erenumab does not have any effect on the human cerebral, meningeal, mammary and coronary arteries, this antibody can antagonize the CGRP-induced vasorelaxation. In this context: (i) CGRP is considered as a protective molecule that could be relevant in some cardiovascular pathologies [110], as suggested in *in vitro* experiments using coronary arteries [111]; and (ii) classical antimigraine agents contraindicated in patients with cardio- and cerebro-vascular risks [31, 53] can also block the CGRP release in the systemic vasculature [68, 112, 113].

Although the short-term tolerability of these mAbs is excellent, theoretical safety concerns should be addressed in phase IV studies [43, 108]. Furthermore, a key issue in the study by Sun *et al.* [101] using erenumab, was that in 10% of patients treated, development of antibodies against the mAb was detected; this finding suggests that under a long-term treatment these antibodies could be an obstacle. Indeed, any kind of mAbs treatment could induce the production of antibodies, which would, in turn, neutralize the action of mAbs; this issue is well illustrated in the case of anti-TNF drugs use [114].

4. PERIPHERAL ACTION OF THE MONOCLONAL ANTIBODIES AGAINST THE CGRPERGIC TRANSMISSION

4.1. Generalities

The mAbs are broken down by renal proteolysis into small peptides and amino acids. Since the metabolism of proteins does not generate toxic intermediary molecules, these compounds are devoid of liver injury [115, 116]. The main pharmacodynamic improvement of these molecules is associated with the high affinity and selectivity against their target, whereas the pharmacokinetic improvement is associated with the long elimination half-life, an absence of metabolism by liver enzymes. In the case of erenumab, this molecule binds to the human CGRP receptor with a high affinity comparable to that of telcagepant, and *in vitro* assays have demonstrated that this mAb inhibits the production of CGRP-induced cAMP, the second messenger associated with the activation of CGRP receptors [117]. On the other hand, in the case of nezumabs, particularly fremanezumab, these compounds seem to act as CGRP scavengers. Indeed, nezumabs inhibit the vasodilator effects induced by capsaicin, a molecule that activates TRPV1 channels, consequently, induces depolarization of perivascular sensory fibers facilitating CGRP release [118].

4.2. Central vs. Peripheral Mechanisms

4.2.1. Generalities on the Peripheral vs. Central Mechanisms in Antimigraine Action

Regarding the general mechanisms involved in acute antimigraine action, the ergots and triptans have, at least, three actions to abort migraine [13, 42]: (i) vasoconstriction of cranial blood vessels; (ii) decrease in CGRP release and other vasoactive neuropeptides; and (iii) inhibition of the trigeminal pain integration. In all cases, these effects are

mediated by direct activation of metabotropic receptors; ergots are less selective and, consequently, induce more adverse events than the triptans [43]. In the case of CGRPergic transmission, Fig. (2) shows that CGRP released from sensory nerves exerts a vasodilator action at the peripheral level by the activation of vascular CGRP receptors; in this case, the classical drugs like ergots and triptans can modulate the perivascular CGRPergic outflow (apart from their direct vasoconstrictor effect). The inhibition of CGRP release is mediated by prejunctional activation of metabotropic receptors coupled to $G_{i/o}$ proteins (*i.e.*, 5-HT_{1B/1D} for the triptans as well as α_2 -adrenoceptors and D₂-like receptors for the ergots) [54]. In this context, the mechanism of action of mAbs to abort migraine attacks may involve: (i) prevention of cranial vasodilatation induced by CGRP; and (ii) blockade of trigeminal pain transmission, at peripheral (trigeminal ganglia) level [53].

4.2.2. The Blood-Brain Barrier in Migraine Pharmacotherapy

After parenteral administration, the mAbs enter vascular endothelial cells by pinocytosis, and due to their nature (*i.e.*, large molecules, ~150 kDa), passage across the Blood-Brain Barrier (BBB) seems to be limited. Indeed, IgGs are general-

ly inaccessible to the brain in part due to the lack of transport mechanisms within the brain capillary wall [119, 120]; hence, the main hypothesis is that these molecules act at the peripheral level, at craniovascular (intra- and extracranial vessels), and also in other neuronal structures relevant to migraine pathophysiology not covered by the BBB (*e.g.*, trigeminal ganglion and the paraventricular structures within the brain stem) [121-123]. This hypothesis is reinforced after considering that the CNS adverse events induced by the mAbs are not different from those of placebo groups.

Since some initial theories suggested that during migraine attacks, the BBB is disrupted, the consensus was that non-lipophilic or high molecular weight compounds can penetrate the CNS exerting their action by central mechanisms. In agreement with this postulate, using experimental models of trigeminovascular neurotransmission, when given parenterally, donitriptan or zolmitriptan (lipophilic), but not sumatriptan (non-lipophilic), inhibited the: (i) *c-fos* expression at TCC induced by electrical stimulation of the Superior Sagittal Sinus (SSS) [124]; and (ii) peripheral capsaicin-induced external carotid vasodilatation [125]. In this sense, sumatriptan was only capable of interrupting the TCC activity when the BBB was disrupted [27] or when given directly at the spinal level [126].

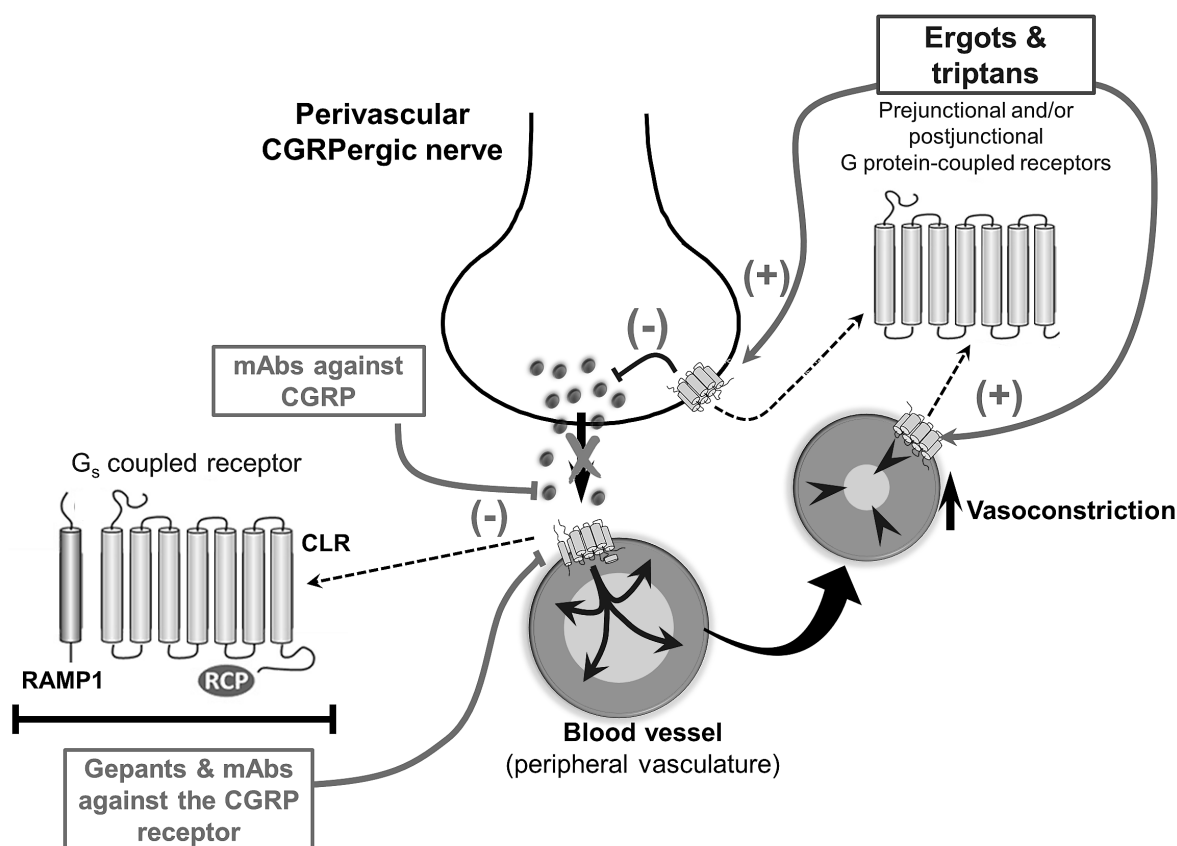


Fig. (2). Main mechanisms involved in the antimigraine action of current specific antimigraine pharmacotherapy. Although not depicted here, similar mechanisms are activated in the case of liposoluble compounds (*i.e.*, they can exert its CGRPergic modulation not only at peripheral, but also at spinal [*i.e.*, on the TCC and supraspinal] levels). In contrast, the antimigraine effects of non-lipophilic compounds and mAbs are restricted to the periphery. Regarding the gepants, although they can cross the BBB, their therapeutic action seems to be restricted to the periphery. Image modified from González-Hernández et al., 2018 [43]. (A higher resolution/colour version of this figure is available in the electronic copy of the article).

More recently, the theory supporting the BBB disruption during migraine has been questioned. Indeed, using the Magnetic Resonance Imaging (MRI) in patients diagnosed with migraine with aura [127] or without aura [128], the integrity of the BBB seems to be intact during migraine attacks. Accordingly, using the Positron-Emission Tomography (PET), an intravenous ^{11}C -dihydroergotamine (at therapeutic doses) injection seems not to cross the BBB in migraine patients [129]. Furthermore, a key study using small CGRP receptor antagonists (gepants), suggested that potential anti-CGRPergic compounds are not required to cross the BBB to exert an antimigraine action. In fact, Hostettler *et al.* [88] showed, by using PET, that telcagepant (given parenterally) did not achieve significant receptor occupancy in the CNS at therapeutic doses, supporting the idea that drugs that act at a peripheral level can produce antimigraine effects. Nevertheless, we must keep in mind the limited spatial resolution of MRI and PET; consequently, we cannot exclude a possible effect of antimigraine compounds in areas closely related to the cerebrospinal fluid not covered by the BBB.

4.2.3. The Antimigraine Action of mAbs can be Explained by their Peripheral Actions

Beyond the fact that CGRP could also be localized in the CNS, at the peripheral level, the meninges are innervated by CGRP immunoreactive fibers [130], and chemical stimulation of these fibers results in the CGRP release [18]. As previously mentioned, mAbs block CGRPergic transmission by blocking the CGRP receptor (numabs) or by acting as CGRP scavengers (nezumabs). In both cases, the mAbs-induced antimigraine action is related to a diminution of CGRPergic function. At this point, taking into account the wide distribution of CGRP and its receptors, we pose the question: where is the main site of action of mAbs for eliciting their therapeutic effect?

Considering only the neuronal theory of migraine, it is tempting to propose that a central action may be indispensable for eliciting a therapeutic antimigraine effect. However, as discussed above, several lines of evidence suggest that central effects are not required for the gepants and mAbs to elicit an antimigraine effect. Indeed, a recent study in rats performed by Nosedá *et al.* [131] using Alexa Fluor 594®-conjugated fremanezumab, showed that after intravenous injection of this fluorescent-labelled mAb, only intense labeling of peripheral tissue was found. Most specifically, dural tissue (including dural blood vessels), trigeminal ganglion, C2 dorsal root ganglion and other autonomic structures related with migraine headaches (*i.e.*, parasympathetic sphenopalatine and the sympathetic superior cervical ganglion) were labeled, whereas no detectable level of mAb was found in the central CGRPergic structures (including the Sp5c, thalamus, hypothalamus and cortex). These findings support the notion that mAbs relieve migraine pain only by a peripheral action on CGRPergic structures.

In contrast, another study using radiolabeled ^{125}I -galcanezumab [132], found not only high levels of this mAb in peripheral tissues, but also a very low level (probably irrelevant to have a therapeutic effect) in the hypothalamus, spinal cord and cerebellum. Certainly, at the hypothalamus, the capillary endothelium is fenestrated [120], allowing the

passage of large proteins. In any case, this study did not find labelling in other central CGRPergic relevant structures.

In further support of the role of a peripheral action to produce an antimigraine effect, a recent study by Edvinsson *et al.* [133] showed that at the peripheral level, complete CGRPergic machinery exists. Briefly, they demonstrated that: (i) at trigeminal ganglion, CGRP is located in C-fibers; (ii) the CLR and RAMP1 (the components of the CGRP receptor) are predominantly found in A δ -fibers; and (iii) the A δ - and C-fibers are in close-contact along the trigeminovascular pathway. Taking together, these findings imply that at trigeminal ganglion, during the activation of C-fibers, CGRP could be released to activate neighboring A δ -fibers containing CGRP receptors (Fig. 3) [133]. This can induce a peripheral sensitization and probably an increase of axonal reflexes mediated by A δ -fibers. Accordingly, compounds with the ability to block peripheral CGRPergic neurotransmission could, in turn, reduce the peripheral sensitization and consequently, the nociceptive input to the TCC (a key site in the pain modulation of orofacial structures).

In earlier full support of this hypothesis, Melo-Carrillo *et al.* [134], using single-unit recording in the trigeminal ganglion, found that parenteral fremanezumab particularly inhibits the activity of meningeal A δ -, but not that of C-fibers. The same authors also showed in animals with a compromised BBB that intravenous fremanezumab does not have any effect on the induction, occurrence, or propagation of Cortical Spreading Depression (CSD) [135]. These results implied that nezumabs (and probably numabs) block preferentially pain signals peripherally and help restore the normal activity (inhibiting the central sensitization) of central structures that receive the peripheral nociceptive input.

4.2.4. The Future in Migraine Pharmacotherapy

Migraine is a neurological disorder where peripheral mechanisms play a key role. Apart from the neuronal effects (central vs. peripheral), the debate on the role of the cranial vasculature in migraine is far from being resolved [23, 36]. In this respect, it is worth noting that current effective specific antimigraine molecules (triptans, gepants and mAbs) reduce dilation of dural arteries. Furthermore, in the case of CGRPergic mAbs, their central actions seem not to be crucial for their therapeutic effect [23, 36]. Nevertheless, the RCT of currently approved mAbs against migraine showed that the overall efficacy of these antibodies does not seem to be dramatic in comparison with the placebo treatment (as pointed out by Pascual [136]), and probably formulations of drugs capable of crossing the BBB could tackle this issue. Certainly, the progress in the field of nanotechnology developing drug delivery systems capable of transporting drugs to the CNS may play a role in future strategies against migraines.

Several studies [137-141] have proposed that nanoparticle formulations could be used with the aim to increase the brain targeting of pharmacological antimigraine drugs. These nanoparticles offer the chance to design smart therapeutic carriers to cross the BBB to the CNS, representing a future perspective to improve the effects of the already existing antimigraine drugs. In this respect, various methodologies have been considered, for example, conjugation of the active

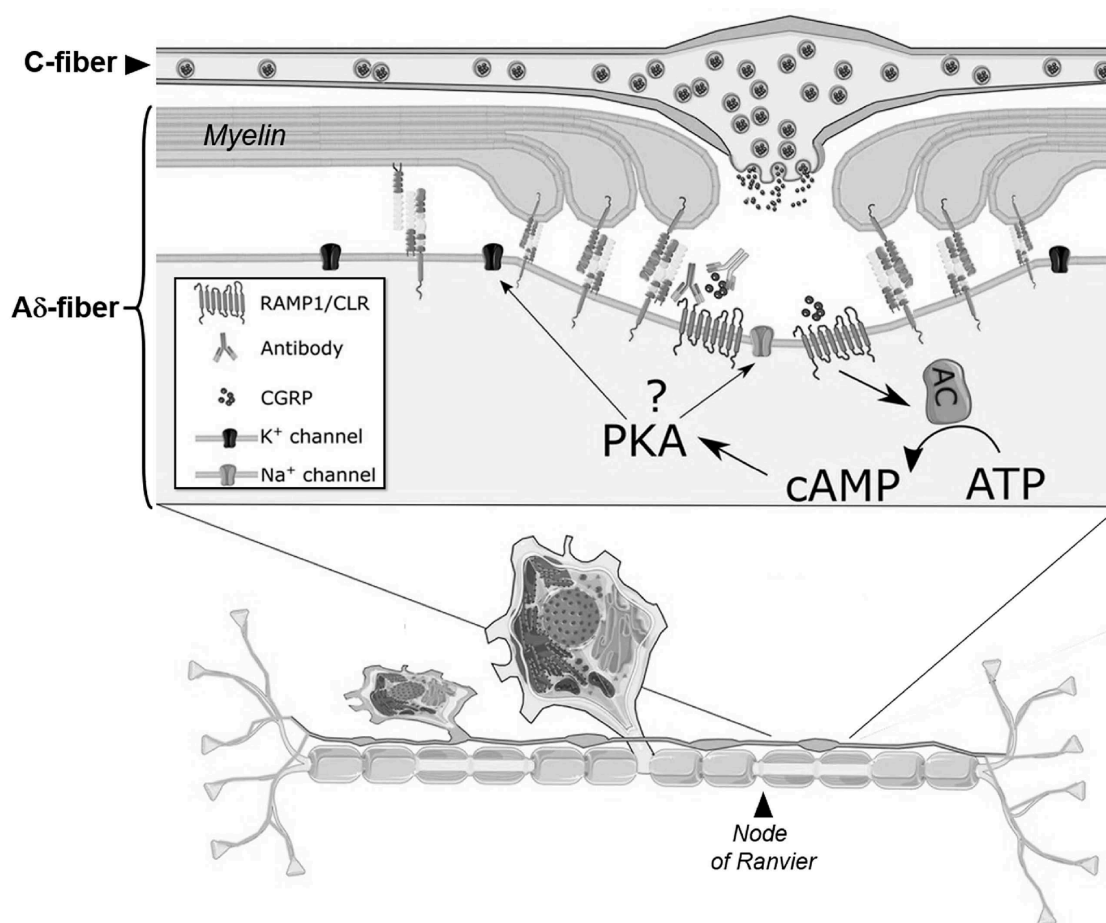


Image modified from Edvinsson et al., 2019 licensed under CC BY 4.0

Fig. (3). Site of action of mAbs at the trigeminovascular peripheral level. At the level of nodes of Ranvier, mAbs can modulate the activity of A δ -fibers (modifying the neuronal properties; e.g., a threshold of activation). CGRP is released from C-fibers in the trigeminal ganglion and regulates A δ -fibers transmission *via* axon-axon signaling. CGRP diffuses into the nodes of Ranvier, it activates the CGRP receptor in A δ -fibers and produces an increase in cAMP, which modifies the activity of Na $^{+}$ or K $^{+}$ channels, probably through PKA. The site of action of mAbs must be targeting CGRP or its receptor in the periphery, most likely in these axo-axonal synapses between C-fibers and A δ -fibers. AC, Adenylyl cyclase; PKA, Protein Kinase. Image modified from Edvinsson et al., 2019 [133]. (A higher resolution/colour version of this figure is available in the electronic copy of the article).

compounds to vectors, nanocarriers (liposomes, polymeric micelles, lipid and polymeric nanoparticles) with high affinity for the BBB [142]. This approach is interesting when considering CGRP receptors, as these receptors are present in the hypothalamus, thalamus, amygdala, periaqueductal gray area, locus coeruleus and raphe magnus, *i.e.*, all areas related with pain modulation and other autonomic/sensory functions [143]. Notwithstanding, as previously discussed, we must keep in mind the CNS adverse events [43]. In addition, we need to underline the unknown effects of long-term treatment with CGRPergic mAbs in the cardiovascular system, since the CGRP receptor is highly expressed in blood vessels [54].

In any case, the current specific antimigraine pharmacotherapy modulating the CGRPergic transmission in the trigeminovascular system (namely, triptans, gepants and mAbs) is not effective in all migraine patients. Accordingly, this fact leads us to hypothesize that, although CGRP plays a role in migraine pathophysiology, several other molecules could also be involved, and perhaps in some patients, the key migraine trigger is not CGRP. Incidentally, the Pituitary Ad-

enylate Cyclase-Activating Peptide (PACAP) and its receptor, the PAC1, are under intense scrutiny as an element involved in migraine pathophysiology [144, 145]. Indeed, a mAb against PACAP (ALD1910) is under Phase I study [146]; nevertheless, in view that PACAP has protective effects in neuronal ischemic injuries, cardiovascular safety studies should be performed [147]. In addition to PACAP, oxytocin and orexin [144, 148-150] may also play a role in migraine pathophysiology, pointing out that the multi-target drug approach could be the next step in the development of successful antimigraine therapies [151].

CONCLUSION

The advent of mAbs against migraine has fostered a discussion on the relevance of peripheral over central mechanisms to achieve an antimigraine effect. The current lines of evidence support the notion that the therapeutic effect of current mAbs is mainly associated with a peripheral action rather than central effects. Indeed, bearing in mind not only the chemical properties of this new class of compounds, but

also the current evidence on the effects of mAbs, the CNS effects seem not to play a major role in the antimigraine effect of these CGRPergic mAbs.

LIST OF ABBREVIATIONS

ALD 403	=	Eptinezumab
AMG-334	=	Erenumab
cAMP	=	3',5'-Cyclic Adenosine Monophosphate
CGRP	=	Calcitonin Gene-Related Peptide
CLR	=	Calcitonin Receptor-Like Receptor
CNS	=	Central Nervous System
EMA	=	European Medicines Agency
FDA	=	U.S. Food and Drug Administration
5-HT	=	5-Hydroxytryptamine or Serotonin
IgG	=	Immunoglobulin Type G
LY2951742	=	Galcanezumab
mAb	=	Monoclonal Antibody
NSAID	=	Non-steroidal Anti-Inflammatory Drug
PACAP	=	Pituitary Adenylate Cyclase-Activating Polypeptide
RAMP1	=	Receptor Activity Modifying Protein 1
RCP	=	Receptor Component Protein
RCT	=	Randomized Clinical Trial
Sp5c	=	Trigeminal Nucleus <i>Pars Caudalis</i>
TCC	=	Trigemino-cervical Complex
TEV-48125	=	Fremanezumab

AUTHORS' CONTRIBUTIONS

All the authors equally wrote, reviewed and approved the final manuscript.

CONSENT FOR PUBLICATION

Not applicable.

FUNDING

This work was financially supported by: (i) the “Fondo Sectorial de Investigación para la Educación” (CONACyT, Grant No. A1-S-23631 to AGH); and (ii) Consejo Nacional de Ciencia y Tecnología (CONACyT, Grant No. 219707 to CMV) and the SEP-Cinvestav Research Support Fund (Grant No. 50 to CMV).

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

ACKNOWLEDGEMENTS

We are grateful to all of our colleagues who collaborated in the studies cited in this review.

REFERENCES

- [1] Akerman S, Holland PR, Goadsby PJ. Diencephalic and brainstem mechanisms in migraine. *Nat Rev Neurosci* 2011; 12(10): 570-84. <http://dx.doi.org/10.1038/nrn3057> PMID: 21931334
- [2] Goadsby PJ, Holland PR, Martins-Oliveira M, Hoffmann J, Schankin C, Akerman S. Pathophysiology of migraine: a disorder of sensory processing. *Physiol Rev* 2017; 97(2): 553-622. <http://dx.doi.org/10.1152/physrev.00034.2015> PMID: 28179394
- [3] Uddman R, Tajti J, Hou M, Sundler F, Edvinsson L. Neuropeptide expression in the human trigeminal nucleus caudalis and in the cervical spinal cord C1 and C2. *Cephalalgia* 2002; 22(2): 112-6. <http://dx.doi.org/10.1046/j.1468-2982.2002.00324.x> PMID: 11972578
- [4] Charles A. Migraine: a brain state. *Curr Opin Neurol* 2013; 26(3): 235-9. <http://dx.doi.org/10.1097/WCO.0b013e32836085f4> PMID: 23493160
- [5] de Tommaso M, Ambrosini A, Brighina F, *et al.* Altered processing of sensory stimuli in patients with migraine. *Nat Rev Neurol* 2014; 10(3): 144-55. <http://dx.doi.org/10.1038/nrneurol.2014.14> PMID: 24535465
- [6] GBD 2016 Neurology Collaborators. Global, regional, and national burden of neurological disorders, 1990-2016: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet Neurol* 2019; 18(5): 459-80. [http://dx.doi.org/10.1016/S1474-4422\(18\)30499-X](http://dx.doi.org/10.1016/S1474-4422(18)30499-X) PMID: 30879893
- [7] GBD 2016 Headache Collaborators. Global, regional, and national burden of migraine and tension-type headache, 1990-2016: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet Neurol* 2018; 17(11): 954-76. [http://dx.doi.org/10.1016/S1474-4422\(18\)30322-3](http://dx.doi.org/10.1016/S1474-4422(18)30322-3) PMID: 30353868
- [8] Tfelt-Hansen P, Lindqvist JK, Do TP. Evaluating the reporting of adverse events in controlled clinical trials conducted in 2010-2015 on migraine drug treatments. *Cephalalgia*, 2018; 38(12): 1885-95. <https://doi.org/10.1177/0333102418759785>
- [9] Villalón CM, Olesen J. The role of CGRP in the pathophysiology of migraine and efficacy of CGRP receptor antagonists as acute antimigraine drugs. *Pharmacol Ther* 2009; 124(3): 309-23. <http://dx.doi.org/10.1016/j.pharmthera.2009.09.003> PMID: 19796656
- [10] Olesen J, Burstein R, Ashina M, Tfelt-Hansen P. Origin of pain in migraine: evidence for peripheral sensitisation. *Lancet Neurol* 2009; 8(7): 679-90. [http://dx.doi.org/10.1016/S1474-4422\(09\)70090-0](http://dx.doi.org/10.1016/S1474-4422(09)70090-0) PMID: 19539239
- [11] Goel D, Un Nisa K, Reza MI, Rahman Z, Aamer S. Aberrant DNA methylation pattern may enhance susceptibility to migraine: A Novel Perspective. *CNS Neurol Disord Drug Targets* 2019; 18(7): 504-15. <http://dx.doi.org/10.2174/1871527318666190809162631> PMID: 31400273
- [12] Villalón CM, VanDenBrink AM. The role of 5-hydroxytryptamine in the pathophysiology of migraine and its relevance to the design of novel treatments. *Mini Rev Med Chem* 2017; 17(11): 928-38. <http://dx.doi.org/10.2174/1389557516666160728121050> PMID: 27465216
- [13] Bernstein C, Burstein R. Sensitization of the trigeminovascular pathway: perspective and implications to migraine pathophysiology. *J Clin Neurol* 2012; 8(2): 89-99. <http://dx.doi.org/10.3988/jcn.2012.8.2.89> PMID: 22787491
- [14] Goadsby PJ, Edvinsson L, Ekman R. Release of vasoactive peptides in the extracerebral circulation of humans and the cat during activation of the trigeminovascular system. *Ann Neurol* 1988; 23(2): 193-6. <http://dx.doi.org/10.1002/ana.410230214> PMID: 2454066
- [15] Goadsby PJ, Hoskin KL. The distribution of trigeminovascular afferents in the nonhuman primate brain *Macaca nemestrina*: a c-fos immunocytochemical study. *J Anat* 1997; 190(Pt 3): 367-75. <http://dx.doi.org/10.1046/j.1469-7580.1997.19030367.x> PMID: 9147223

- [16] Bartsch T, Goadsby PJ. Stimulation of the greater occipital nerve induces increased central excitability of dural afferent input. *Brain* 2002; 125(Pt 7): 1496-509.
<http://dx.doi.org/10.1093/brain/awf166> PMID: 12077000
- [17] Penfield W, McNaughton FL. Dural headache and the innervation of the dura mater. *Arch Neurol Psychiatry* 1940; 44: 43-75.
<http://dx.doi.org/10.1001/archneurpsyc.1940.02280070051003>
- [18] Ebersberger A, Aeverbeck B, Messlinger K, Reeh PW. Release of substance P, calcitonin gene-related peptide and prostaglandin E2 from rat dura mater encephali following electrical and chemical stimulation *in vitro*. *Neuroscience* 1999; 89(3): 901-7.
[http://dx.doi.org/10.1016/S0306-4522\(98\)00366-2](http://dx.doi.org/10.1016/S0306-4522(98)00366-2) PMID: 10199623
- [19] Williamson DJ, Hargreaves RJ, Hill RG, Shepherd SL. Sumatriptan inhibits neurogenic vasodilation of dural blood vessels in the anaesthetized rat--intravital microscope studies. *Cephalalgia* 1997; 17(4): 525-31.
<http://dx.doi.org/10.1046/j.1468-2982.1997.1704525.x> PMID: 9209774
- [20] Humphrey PP, Feniuk W, Perren MJ, Beresford IJ, Skingle M, Whalley ET. Serotonin and migraine. *Ann N Y Acad Sci* 1990; 600: 587-98.
<http://dx.doi.org/10.1111/j.1749-6632.1990.tb16912.x> PMID: 2252337
- [21] Goadsby PJ. The vascular theory of migraine--a great story wrecked by the facts. *Brain* 2009; 132(Pt 1): 6-7.
<http://dx.doi.org/10.1093/brain/awn321> PMID: 19098031
- [22] Mason BN, Russo AF. Vascular Contributions to Migraine: Time to Revisit? *Front Cell Neurosci* 2018; 12: 233.
<http://dx.doi.org/10.3389/fncel.2018.00233> PMID: 30127722
- [23] Jacobs B, Dussor G. Neurovascular contributions to migraine: Moving beyond vasodilation. *Neuroscience* 2016; 338: 130-44.
<http://dx.doi.org/10.1016/j.neuroscience.2016.06.012> PMID: 27312704
- [24] Kayser V, Aubel B, Hamon M, Bourgoin S. The antimigraine 5-HT 1B/1D receptor agonists, sumatriptan, zolmitriptan and dihydroergotamine, attenuate pain-related behaviour in a rat model of trigeminal neuropathic pain. *Br J Pharmacol* 2002; 137(8): 1287-97.
<http://dx.doi.org/10.1038/sj.bjp.0704979> PMID: 12466238
- [25] Wolff HG. Headache and other Head Pain. New York: Oxford Univ. Press 1963.
- [26] Feniuk W, Humphrey PPA, Perren MJ, Connor HE, Whalley ET. Rationale for the use of 5-HT₁-like agonists in the treatment of migraine. *J Neurol* 1991; 238(Suppl. 1): S57-61.
<http://dx.doi.org/10.1007/BF01642908> PMID: 1646289
- [27] Kaube H, Hoskin KL, Goadsby PJ. Inhibition by sumatriptan of central trigeminal neurones only after blood-brain barrier disruption. *Br J Pharmacol* 1993; 109(3): 788-92.
<http://dx.doi.org/10.1111/j.1476-5381.1993.tb13643.x> PMID: 8395298
- [28] Humphrey PPA, Goadsby PJ. The mode of action of sumatriptan is vascular? A debate. *Cephalalgia* 1994; 14(6): 401-10.
<http://dx.doi.org/10.1046/j.1468-2982.1994.1406401.x> PMID: 7697699
- [29] Shevel E. The extracranial vascular theory of migraine--a great story confirmed by the facts. *Headache* 2011; 51(3): 409-17.
<http://dx.doi.org/10.1111/j.1526-4610.2011.01844.x> PMID: 21352215
- [30] Shevel E. The extracranial vascular theory of migraine: an artificial controversy. *J Neural Transm (Vienna)* 2011; 118(4): 525-30.
<http://dx.doi.org/10.1007/s00702-010-0517-1> PMID: 21207080
- [31] Messlinger K, MaassenVanDenBrink A. Cardio- and cerebrovascular safety of erenumab, a monoclonal antibody targeting CGRP receptors--important studies on human isolated arteries. *Cephalalgia* 2019; 39: 1731-4.
<http://dx.doi.org/10.1177/0333102419877169> PMID: 31526030
- [32] Goadsby PJ. Pathophysiology of migraine. *Neurol Clin* 2009; 27(2): 335-60.
<http://dx.doi.org/10.1016/j.ncl.2008.11.012> PMID: 19289219
- [33] Shevel E. Comments on Schoonman *et al.*: Migraine headache is not associated with cerebral or meningeal vasodilatation: a 3 T magnetic resonance angiography study (*Brain* 2008; 131:2192-2200). *J Headache Pain* 2009; 10(4): 307-8.
<http://dx.doi.org/10.1007/s10194-009-0123-5> PMID: 19407929
- [34] Drummond PD, Lance JW. Extracranial vascular changes and the source of pain in migraine headache. *Ann Neurol* 1983; 13(1): 32-7.
<http://dx.doi.org/10.1002/ana.410130108> PMID: 6830162
- [35] Hoffmann J, Baca SM, Akerman S. Neurovascular mechanisms of migraine and cluster headache. *J Cereb Blood Flow Metab* 2019; 39(4): 573-94.
<http://dx.doi.org/10.1177/0271678X17733655> PMID: 28948863
- [36] Dodick DW. Examining the essence of migraine--is it the blood vessel or the brain? A debate. *Headache* 2008; 48(4): 661-7.
<http://dx.doi.org/10.1111/j.1526-4610.2008.01079.x> PMID: 18377395
- [37] Frederiksen SD, Haanes KA, Warfvinge K, Edvinsson L. Perivascular neurotransmitters: Regulation of cerebral blood flow and role in primary headaches. *J Cereb Blood Flow Metab* 2019; 39(4): 610-32.
<http://dx.doi.org/10.1177/0271678X17747188> PMID: 29251523
- [38] Strassman AM, Raymond SA, Burstein R. Sensitization of meningeal sensory neurons and the origin of headaches. *Nature* 1996; 384(6609): 560-4.
<http://dx.doi.org/10.1038/384560a0> PMID: 8955268
- [39] Burstein R, Yamamura H, Malick A, Strassman AM. Chemical stimulation of the intracranial dura induces enhanced responses to facial stimulation in brain stem trigeminal neurons. *J Neurophysiol* 1998; 79(2): 964-82.
<http://dx.doi.org/10.1152/jn.1998.79.2.964> PMID: 9463456
- [40] Burstein R, Cutrer MF, Yarnitsky D. The development of cutaneous allodynia during a migraine attack clinical evidence for the sequential recruitment of spinal and supraspinal nociceptive neurons in migraine. *Brain* 2000; 123(Pt 8): 1703-9.
<http://dx.doi.org/10.1093/brain/123.8.1703> PMID: 10908199
- [41] Ashina M, Hansen JM, Do TP, Melo-Carrillo A, Burstein R, Moskowitz MA. Migraine and the trigeminovascular system--40 years and counting. *Lancet Neurol* 2019; 18(8): 795-804.
[http://dx.doi.org/10.1016/S1474-4422\(19\)30185-1](http://dx.doi.org/10.1016/S1474-4422(19)30185-1) PMID: 31160203
- [42] Tfelt-Hansen P, De Vries P, Saxena PR. Triptans in migraine: a comparative review of pharmacology, pharmacokinetics and efficacy. *Drugs* 2000; 60(6): 1259-87.
<http://dx.doi.org/10.2165/00003495-200060060-00003> PMID: 11152011
- [43] González-Hernández A, Marichal-Cancino BA, MaassenVanDenBrink A, Villalón CM. Side effects associated with current and prospective antimigraine pharmacotherapies. *Expert Opin Drug Metab Toxicol* 2018; 14(1): 25-41.
<http://dx.doi.org/10.1080/17425255.2018.1416097> PMID: 29226741
- [44] Rubio-Beltrán E, Labastida-Ramírez A, Villalón CM, Maassen-VanDenBrink A. Is selective 5-HT_{1F} receptor agonism an entity apart from that of the triptans in antimigraine therapy? *Pharmacol Ther* 2018; 186: 88-97.
<http://dx.doi.org/10.1016/j.pharmthera.2018.01.005> PMID: 29352859
- [45] Goadsby PJ. The pharmacology of headache. *Prog Neurobiol* 2000; 62(5): 509-25.
[http://dx.doi.org/10.1016/S0301-0082\(00\)00010-1](http://dx.doi.org/10.1016/S0301-0082(00)00010-1) PMID: 10869781
- [46] Bouchelet I, Case B, Olivier A, Hamel E. No contractile effect for 5-HT_{1D} and 5-HT_{1F} receptor agonists in human and bovine cerebral arteries: similarity with human coronary artery. *Br J Pharmacol* 2000; 129(3): 501-8.
<http://dx.doi.org/10.1038/sj.bjp.0703081> PMID: 10711348
- [47] Färkkilä M, Diener HC, Géraud G, *et al.* COL MIG-202 study group. Efficacy and tolerability of lasmiditan, an oral 5-HT_{1F} receptor agonist, for the acute treatment of migraine: a phase 2 randomised, placebo-controlled, parallel-group, dose-ranging study. *Lancet Neurol* 2012; 11(5): 405-13.
[http://dx.doi.org/10.1016/S1474-4422\(12\)70047-9](http://dx.doi.org/10.1016/S1474-4422(12)70047-9) PMID: 22459549
- [48] Johnson KW, Schaus JM, Durkin MM, *et al.* 5-HT_{1F} receptor agonists inhibit neurogenic dural inflammation in guinea pigs. *Neuroreport* 1997; 8(9-10): 2237-40.
<http://dx.doi.org/10.1097/00001756-199707070-00029> PMID: 9243618

- [49] Mitsikostas DD, Sanchez del Rio M, Waeber C. 5-Hydroxytryptamine(1B/1D) and 5-hydroxytryptamine1F receptors inhibit capsaicin-induced c-fos immunoreactivity within mouse trigeminal nucleus caudalis. *Cephalalgia* 2002; 22(5): 384-94. <http://dx.doi.org/10.1046/j.1468-2982.2002.00382.x> PMID: 12110114
- [50] Nelson DL, Phebus LA, Johnson KW, *et al.* Preclinical pharmacological profile of the selective 5-HT_{1F} receptor agonist lasmiditan. *Cephalalgia* 2010; 30(10): 1159-69. <http://dx.doi.org/10.1177/0333102410370873> PMID: 20855361
- [51] Phebus LA, Johnson KW, Zgombick JM, *et al.* Characterization of LY344864 as a pharmacological tool to study 5-HT_{1F} receptors: binding affinities, brain penetration and activity in the neurogenic dural inflammation model of migraine. *Life Sci* 1997; 61(21): 2117-26. [http://dx.doi.org/10.1016/S0024-3205\(97\)00885-0](http://dx.doi.org/10.1016/S0024-3205(97)00885-0) PMID: 9395253
- [52] Moreno-Ajona D, Chan C, Villar-Martinez MD, Goadsby PJ. Targeting CGRP and 5-HT_{1F} Receptors for the Acute Therapy of Migraine: A Literature Review. *Headache* 2019; 59(Suppl. 2): 3-19. <http://dx.doi.org/10.1111/head.13582> PMID: 31291016
- [53] MaassenVanDenBrink A, Meijer J, Villalón CM, Ferrari MD. Wiping out CGRP: potential cardiovascular risks. *Trends Pharmacol Sci* 2016; 37: 779-88. <http://dx.doi.org/10.1016/j.tips.2016.06.002>
- [54] González-Hernández A, Marichal-Cancino BA, Lozano-Cuenca J, *et al.* Heteroreceptors modulating CGRP release at neurovascular junction: potential therapeutic implications on some vascular-related diseases. *BioMed Res Int* 2016; 20162056786 <http://dx.doi.org/10.1155/2016/2056786> PMID: 28116293
- [55] Amrutkar DV, Ploug KB, Hay-Schmidt A, Porreca F, Olesen J, Jansen-Olesen I. mRNA expression of 5-hydroxytryptamine 1B, 1D, and 1F receptors and their role in controlling the release of calcitonin gene-related peptide in the rat trigeminovascular system. *Pain* 2012; 153(4): 830-8. <http://dx.doi.org/10.1016/j.pain.2012.01.005> PMID: 22305629
- [56] Goadsby PJ, Edvinsson L. The trigeminovascular system and migraine: studies characterizing cerebrovascular and neuropeptide changes seen in humans and cats. *Ann Neurol* 1993; 33(1): 48-56. <http://dx.doi.org/10.1002/ana.410330109> PMID: 8388188
- [57] Tso AR, Goadsby PJ. Anti-CGRP monoclonal antibodies: the next era of migraine prevention? *Curr Treat Options Neurol* 2017; 19(8): 27. <http://dx.doi.org/10.1007/s11940-017-0463-4> PMID: 28653227
- [58] Mitsikostas DD, Rapoport AM. New players in the preventive treatment of migraine. *BMC Med* 2015; 13: 279. <http://dx.doi.org/10.1186/s12916-015-0522-1> PMID: 26555040
- [59] Diener HC, Charles A, Goadsby PJ, Holle D. New therapeutic approaches for the prevention and treatment of migraine. *Lancet Neurol* 2015; 14(10): 1010-22. [http://dx.doi.org/10.1016/S1474-4422\(15\)00198-2](http://dx.doi.org/10.1016/S1474-4422(15)00198-2) PMID: 26376968
- [60] Charles A, Pozo-Rosich P. Targeting calcitonin gene-related peptide: a new era in migraine therapy. *Lancet* 2019; 394(10210): 1765-74. [http://dx.doi.org/10.1016/S0140-6736\(19\)32504-8](http://dx.doi.org/10.1016/S0140-6736(19)32504-8) PMID: 31668411
- [61] Xu H, Han W, Wang J, Li M. Network meta-analysis of migraine disorder treatment by NSAIDs and triptans. *J Headache Pain* 2016; 17(1): 113. <http://dx.doi.org/10.1186/s10194-016-0703-0> PMID: 27957624
- [62] Ong JJY, De Felice M. Migraine treatment: Current acute medications and their potential mechanisms of action. *Neurotherapeutics* 2018; 15(2): 274-90. <http://dx.doi.org/10.1007/s13311-017-0592-1> PMID: 29235068
- [63] Silberstein SD. The pharmacology of ergotamine and dihydroergotamine. *Headache* 1997; 37(Suppl. 1): S15-25. PMID: 9009470
- [64] Rothlin E. The specific action of ergot alkaloids on the sympathetic nervous system. *J Pharmacol Exp Ther* 1929; 36: 657-83.
- [65] Garrett WJ. The effects of adrenaline noradrenaline and dihydroergotamine on excised human myometrium. *Br J Pharmacol Chemother* 1955; 10(1): 39-44. <http://dx.doi.org/10.1111/j.1476-5381.1955.tb00057.x> PMID: 14351684
- [66] Lennox WG. The Use of Ergotamine Tartrate in Migraine. *N Engl J Med* 1934; 210: 1061-5. <http://dx.doi.org/10.1056/NEJM193405172102004>
- [67] Lennox WG, Von Storch TJC. Experience with ergotamine tartrate in 120 patients with migraine. *JAMA* 1935; 105: 169-71. <http://dx.doi.org/10.1001/jama.1935.02760290003002>
- [68] González-Hernández A, Marichal-Cancino BA, Lozano-Cuenca J, MaassenVanDenBrink A, Villalón CM. MaassenVanDenBrink A, Villalón CM. Functional characterization of the prejunctional receptors mediating the inhibition by ergotamine of the rat perivascular sensory peptidergic drive. *ACS Chem Neurosci* 2019; 10(7): 3173-82. <http://dx.doi.org/10.1021/acscchemneuro.8b00611> PMID: 30695640
- [69] Chan KY, Vermeersch S, de Hoon J, Villalón CM, Maassen-vandenbrink A. Potential mechanisms of prospective antimigraine drugs: a focus on vascular (side) effects. *Pharmacol Ther* 2011; 129(3): 332-51. <http://dx.doi.org/10.1016/j.pharmthera.2010.12.001> PMID: 21130807
- [70] Soveyd N, Abdolahi M, Djalali M, *et al.* The combined effects of ω -3 fatty acids and nano-curcumin supplementation on intercellular adhesion molecule-1 (ICAM-1) gene expression and serum levels in migraine patients. *CNS Neurol Disord Drug Targets* 2018; 16(10): 1120-6. <http://dx.doi.org/10.2174/1871527317666171213154749> PMID: 29237386
- [71] Abdolahi M, Sarraf P, Javanbakht MH, *et al.* A Novel combination of ω -3 fatty acids and nano-curcumin modulates interleukin-6 gene expression and high sensitivity c-reactive protein serum levels in patients with migraine: a randomized clinical trial study. *CNS Neurol Disord Drug Targets* 2018; 17(6): 430-8. <http://dx.doi.org/10.2174/1871527317666180625101643> PMID: 29938621
- [72] Abdolahi M, Tafakhori A, Togha M, *et al.* The synergistic effects of ω -3 fatty acids and nano-curcumin supplementation on tumor necrosis factor (TNF)- α gene expression and serum level in migraine patients. *Immunogenetics* 2017; 69(6): 371-8. <http://dx.doi.org/10.1007/s00251-017-0992-8> PMID: 28478481
- [73] Villalón CM, Centurión D, Valdivia LF, De Vries P, Saxena PR. An introduction to migraine: from ancient treatment to functional pharmacology and antimigraine therapy. *Proc West Pharmacol Soc* 2002; 45: 199-210. PMID: 12434581
- [74] Dodick DW, Martin VT, Smith T, Silberstein S. Cardiovascular tolerability and safety of triptans: a review of clinical data. *Headache* 2004; 44(S1)(Suppl. 1): S20-30. <http://dx.doi.org/10.1111/j.1526-4610.2004.04105.x> PMID: 15149490
- [75] Doenicke A, Brand J, Perrin VL. Possible benefit of GR43175, a novel 5-HT₁-like receptor agonist, for the acute treatment of severe migraine. *Lancet* 1988; 1(8598): 1309-11. [http://dx.doi.org/10.1016/S0140-6736\(88\)92122-8](http://dx.doi.org/10.1016/S0140-6736(88)92122-8) PMID: 2897560
- [76] Patten JP. Oral Sumatriptan Dose-defining Study Group. Clinical experience with oral sumatriptan: a placebo-controlled, dose-ranging study. *J Neurol* 1991; 238(S1)(Suppl. 1): S62-5. <http://dx.doi.org/10.1007/BF01642909> PMID: 1646290
- [77] Ferrari MD, Goadsby PJ, Roon KI, Lipton RB. Triptans (serotonin, 5-HT_{1B/1D} agonists) in migraine: detailed results and methods of a meta-analysis of 53 trials. *Cephalalgia* 2002; 22(8): 633-58. <http://dx.doi.org/10.1046/j.1468-2982.2002.00404.x> PMID: 12383060
- [78] Pini LA, Brovia D. Different characteristics of triptans. *J Headache Pain* 2004; 5: S109-11. <http://dx.doi.org/10.1007/s10194-004-0122-5>
- [79] Diener HC, Limmroth V. Advances in pharmacological treatment of migraine. *Expert Opin Investig Drugs* 2001; 10(10): 1831-45. <http://dx.doi.org/10.1517/13543784.10.10.1831> PMID: 11772289
- [80] Burger-Mulder I, Li M, Quin T, *et al.* Anti-migraine CGRP antagonists (gepants) worsen cerebral ischemia outcome in mice. *Cephalalgia* 2019; 39(1S): 398.
- [81] Do TP, Guo S, Ashina M. Therapeutic novelties in migraine: new drugs, new hope? *J Headache Pain* 2019; 20(1): 37. <http://dx.doi.org/10.1186/s10194-019-0974-3> PMID: 30995909

- [82] Olesen J, Diener HC, Husstedt IW, *et al.* BIBN 4096 BS Clinical Proof of Concept Study Group. Calcitonin gene-related peptide receptor antagonist BIBN 4096 BS for the acute treatment of migraine. *N Engl J Med* 2004; 350(11): 1104-10.
<http://dx.doi.org/10.1056/NEJMoa030505> PMID: 15014183
- [83] Ho TW, Mannix LK, Fan X, *et al.* MK-0974 Protocol 004 study group. Randomized controlled trial of an oral CGRP receptor antagonist, MK-0974, in acute treatment of migraine. *Neurology* 2008; 70(16): 1304-12.
<http://dx.doi.org/10.1212/01.WNL.0000286940.29755.61> PMID: 17914062
- [84] Voss T, Lipton RB, Dodick DW, *et al.* A phase IIb randomized, double-blind, placebo-controlled trial of ubrogepant for the acute treatment of migraine. *Cephalalgia* 2016; 36(9): 887-98.
<http://dx.doi.org/10.1177/0333102416653233> PMID: 27269043
- [85] Tepper SJ. Anti-calcitonin gene-related peptide (CGRP) therapies: Update on a previous review after the American Headache Society 60th Scientific Meeting, San Francisco, June 2018. *Headache* 2018; 58(Suppl. 3): 276-90.
<http://dx.doi.org/10.1111/head.13417> PMID: 30403405
- [86] Negro A, Martelletti P. Gepants for the treatment of migraine. *Expert Opin Investig Drugs* 2019; 28(6): 555-67.
<http://dx.doi.org/10.1080/13543784.2019.1618830> PMID: 31081399
- [87] Martelletti P, Giamberardino MA. Advances in orally administered pharmacotherapy for the treatment of migraine. *Expert Opin Pharmacother* 2019; 20(2): 209-18.
<http://dx.doi.org/10.1080/14656566.2018.1549223> PMID: 30475090
- [88] Hostetler ED, Joshi AD, Sanabria-Bohórquez S, *et al.* *In vivo* quantification of calcitonin gene-related peptide receptor occupancy by telcagepant in rhesus monkey and human brain using the positron emission tomography tracer [¹¹C]MK-4232. *J Pharmacol Exp Ther* 2013; 347(2): 478-86.
<http://dx.doi.org/10.1124/jpet.113.206458> PMID: 23975906
- [89] Dodick DW, Goadsby PJ, Silberstein SD, *et al.* ALD403 study investigators. Safety and efficacy of ALD403, an antibody to calcitonin gene-related peptide, for the prevention of frequent episodic migraine: a randomised, double-blind, placebo-controlled, exploratory phase 2 trial. *Lancet Neurol* 2014; 13(11): 1100-7.
[http://dx.doi.org/10.1016/S1474-4422\(14\)70209-1](http://dx.doi.org/10.1016/S1474-4422(14)70209-1) PMID: 25297013
- [90] Bigal ME, Edvinsson L, Rapoport AM, *et al.* Safety, tolerability, and efficacy of TEV-48125 for preventive treatment of chronic migraine: a multicentre, randomised, double-blind, placebo-controlled, phase 2b study. *Lancet Neurol* 2015; 14(11): 1091-100.
[http://dx.doi.org/10.1016/S1474-4422\(15\)00245-8](http://dx.doi.org/10.1016/S1474-4422(15)00245-8) PMID: 26432181
- [91] Dodick DW, Goadsby PJ, Spierings EL, Scherer JC, Sweeney SP, Grayzel DS. Safety and efficacy of LY2951742, a monoclonal antibody to calcitonin gene-related peptide, for the prevention of migraine: a phase 2, randomised, double-blind, placebo-controlled study. *Lancet Neurol* 2014; 13(9): 885-92.
[http://dx.doi.org/10.1016/S1474-4422\(14\)70128-0](http://dx.doi.org/10.1016/S1474-4422(14)70128-0) PMID: 25127173
- [92] Detke HC, Goadsby PJ, Wang S, Friedman DI, Selzler KJ, Aurora SK. Galcanezumab in chronic migraine: The randomized, double-blind, placebo-controlled REGAIN study. *Neurology* 2018; 91(24): e2211-21.
<http://dx.doi.org/10.1212/WNL.0000000000006640> PMID: 30446596
- [93] Skljarevski V, Oakes TM, Zhang Q, *et al.* Effect of Different Doses of Galcanezumab vs Placebo for Episodic Migraine Prevention: A Randomized Clinical Trial. *JAMA Neurol* 2018; 75(2): 187-93.
<http://dx.doi.org/10.1001/jamaneurol.2017.3859> PMID: 29255900
- [94] Camporeale A, Kudrow D, Sides R, *et al.* A phase 3, long-term, open-label safety study of Galcanezumab in patients with migraine. *BMC Neurol* 2018; 18(1): 188.
<http://dx.doi.org/10.1186/s12883-018-1193-2> PMID: 30413151
- [95] Holzmann B. Modulation of immune responses by the neuropeptide CGRP. *Amino Acids* 2013; 45(1): 1-7.
<http://dx.doi.org/10.1007/s00726-011-1161-2> PMID: 22113645
- [96] Holzmann B. Antiinflammatory activities of CGRP modulating innate immune responses in health and disease. *Curr Protein Pept Sci* 2013; 14(4): 268-74.
<http://dx.doi.org/10.2174/13892037113149990046> PMID: 23745695
- [97] Rochlitz S, Veres TZ, Kühne K, *et al.* The neuropeptide calcitonin gene-related peptide affects allergic airway inflammation by modulating dendritic cell function. *Clin Exp Allergy* 2011; 41(11): 1609-21.
<http://dx.doi.org/10.1111/j.1365-2222.2011.03822.x> PMID: 21752117
- [98] Overeem LH, Neeb L, Reuter U. Erenumab for episodic migraine prophylaxis. *Expert Rev Neurother* 2019; 19(8): 751-7.
<http://dx.doi.org/10.1080/14737175.2019.1565996> PMID: 30614741
- [99] Tepper S, Ashina M, Reuter U, *et al.* Safety and efficacy of erenumab for preventive treatment of chronic migraine: a randomised, double-blind, placebo-controlled phase 2 trial. *Lancet Neurol* 2017; 16(6): 425-34.
[http://dx.doi.org/10.1016/S1474-4422\(17\)30083-2](http://dx.doi.org/10.1016/S1474-4422(17)30083-2) PMID: 28460892
- [100] Reuter U, Goadsby PJ, Lanteri-Minet M, *et al.* Efficacy and tolerability of erenumab in patients with episodic migraine in whom two-to-four previous preventive treatments were unsuccessful: a randomised, double-blind, placebo-controlled, phase 3b study. *Lancet* 2018; 392(10161): 2280-7.
[http://dx.doi.org/10.1016/S0140-6736\(18\)32534-0](http://dx.doi.org/10.1016/S0140-6736(18)32534-0) PMID: 30360965
- [101] Sun H, Dodick DW, Silberstein S, *et al.* Safety and efficacy of AMG 334 for prevention of episodic migraine: a randomised, double-blind, placebo-controlled, phase 2 trial. *Lancet Neurol* 2016; 15(4): 382-90.
[http://dx.doi.org/10.1016/S1474-4422\(16\)00019-3](http://dx.doi.org/10.1016/S1474-4422(16)00019-3) PMID: 26879279
- [102] Goadsby PJ, Reuter U, Hallström Y, *et al.* A controlled trial of erenumab for episodic migraine. *N Engl J Med* 2017; 377(22): 2123-32.
<http://dx.doi.org/10.1056/NEJMoa1705848> PMID: 29171821
- [103] Raffaelli B, Mussetto V, Israel H, Neeb L, Reuter U. Erenumab and galcanezumab in chronic migraine prevention: effects after treatment termination. *J Headache Pain* 2019; 20(1): 66.
<http://dx.doi.org/10.1186/s10194-019-1018-8> PMID: 31159727
- [104] Mathew PG, Klein BC. Getting to the heart of the matter: migraine, triptans, DHE, ditans, CGRP antibodies, first/second-generation gepants, and cardiovascular risk. *Headache* 2019; 59(8): 1421-6.
<http://dx.doi.org/10.1111/head.13601> PMID: 31318457
- [105] Raffaelli B, Reuter U. The biology of monoclonal antibodies: Focus on calcitonin gene-related peptide for prophylactic migraine therapy. *Neurotherapeutics* 2018; 15(2): 324-35.
<http://dx.doi.org/10.1007/s13311-018-0622-7> PMID: 29616494
- [106] Dodick DW. CGRP ligand and receptor monoclonal antibodies for migraine prevention: Evidence review and clinical implications. *Cephalalgia* 2019; 39(3): 445-58.
<http://dx.doi.org/10.1177/0333102418821662> PMID: 30661365
- [107] Depre C, Antalík L, Starling A, *et al.* A randomized, double-blind, placebo-controlled study to evaluate the effect of erenumab on exercise time during a treadmill test in patients with stable angina. *Headache* 2018; 58(5): 715-23.
<http://dx.doi.org/10.1111/head.13316> PMID: 29878340
- [108] Maassen van den Brink A, Rubio-Beltrán E, Duncker D, Villalón CM. Is CGRP receptor blockade cardiovascularly safe? Appropriate studies are needed. *Headache* 2018; 58(8): 1257-8.
<http://dx.doi.org/10.1111/head.13402> PMID: 30289181
- [109] Denekas T, Tröltzsch M, Vater A, Klusmann S, Messlinger K. Inhibition of stimulated meningeal blood flow by a calcitonin gene-related peptide binding mirror-image RNA oligonucleotide. *Br J Pharmacol* 2006; 148(4): 536-43.
<http://dx.doi.org/10.1038/sj.bjp.0706742> PMID: 16633354
- [110] Kumar A, Potts JD, DiPette DJ. Protective role of α -calcitonin gene-related peptide in cardiovascular diseases. *Front Physiol* 2019; 10: 821.
<http://dx.doi.org/10.3389/fphys.2019.00821> PMID: 31312143
- [111] Homma S, Kimura T, Sakai S, *et al.* Calcitonin gene-related peptide protects the myocardium from ischemia induced by endothelin-

- 1: intravital microscopic observation and (31)P-MR spectroscopic studies. *Life Sci* 2014; 118(2): 248-54.
<http://dx.doi.org/10.1016/j.lfs.2014.02.024> PMID: 24607775
- [112] Lozano-Cuenca J, González-Hernández A, Muñoz-Islas E, *et al.* Effect of some acute and prophylactic antimigraine drugs on the vasodepressor sensory CGRPergic outflow in pithed rats. *Life Sci* 2009; 84(5-6): 125-31.
<http://dx.doi.org/10.1016/j.lfs.2008.11.008> PMID: 19041880
- [113] González-Hernández A, Manrique-Maldonado G, Lozano-Cuenca J, *et al.* The 5-HT(1) receptors inhibiting the rat vasodepressor sensory CGRPergic outflow: further involvement of 5-HT(1F), but not 5-HT(1A) or 5-HT(1D), subtypes. *Eur J Pharmacol* 2011; 659(2-3): 233-43.
<http://dx.doi.org/10.1016/j.ejphar.2011.03.035> PMID: 21473863
- [114] Prado MS, Bendtzen K, Andrade LEC. Biological anti-TNF drugs: immunogenicity underlying treatment failure and adverse events. *Expert Opin Drug Metab Toxicol* 2017; 13(9): 985-95.
<http://dx.doi.org/10.1080/17425255.2017.1360280> PMID: 28772079
- [115] 115. LiverTox: Clinical and Research Information on Drug-Induced Liver Injury Bethesda (MD): National Institute of Diabetes and Digestive and Kidney Diseases 2012.
<https://www.ncbi.nlm.nih.gov/books/NBK548844/>
- [116] Taylor FR. CGRP, amylin, immunology, and headache medicine. *Headache* 2019; 59(1): 131-50.
<http://dx.doi.org/10.1111/head.13432> PMID: 30390312
- [117] Vu T, Ma P, Chen JS, *et al.* Pharmacokinetic pharmacodynamic relationship of erenumab (AMG 334) and capsaicin-induced dermal blood flow in healthy and migraine subjects. *Pharm Res* 2017; 34(9): 1784-95.
<http://dx.doi.org/10.1007/s11095-017-2183-6> PMID: 28593473
- [118] Grell AS, Haanes KA, Johansson SE, Edvinsson L, Sams A. Fremanezumab inhibits vasodilatory effects of CGRP and capsaicin in rat cerebral artery - Potential role in conditions of severe vasoconstriction. *Eur J Pharmacol* 2019; 864172726
<http://dx.doi.org/10.1016/j.ejphar.2019.172726> PMID: 31589869
- [119] Bullard DE, Bourdon M, Bigner DD. Comparison of various methods for delivering radiolabeled monoclonal antibody to normal rat brain. *J Neurosurg* 1984; 61(5): 901-11.
<http://dx.doi.org/10.3171/jns.1984.61.5.0901> PMID: 6436447
- [120] Brightman MW. Morphology of blood-brain interfaces. *Exp Eye Res* 1977; 25(Suppl.): 1-25.
[http://dx.doi.org/10.1016/S0014-4835\(77\)80008-0](http://dx.doi.org/10.1016/S0014-4835(77)80008-0) PMID: 73470
- [121] Edvinsson L. The Trigemino-vascular pathway: role of CGRP and CGRP receptors in migraine. *Headache* 2017; 57(Suppl. 2): 47-55.
<http://dx.doi.org/10.1111/head.13081> PMID: 28485848
- [122] Edvinsson L, Tfelt-Hansen P. The blood-brain barrier in migraine treatment. *Cephalalgia* 2008; 28(12): 1245-58.
<http://dx.doi.org/10.1111/j.1468-2982.2008.01675.x> PMID: 18727638
- [123] Eftekhari S, Salvatore CA, Johansson S, Chen T-B, Zeng Z, Edvinsson L. Localization of CGRP, CGRP receptor, PACAP and glutamate in trigeminal ganglion. Relation to the blood-brain barrier. *Brain Res* 2015; 1600: 93-109.
<http://dx.doi.org/10.1016/j.brainres.2014.11.031> PMID: 25463029
- [124] Hoskin KL, Goadsby PJ. Comparison of more and less lipophilic serotonin (5HT1B/1D) agonists in a model of trigemino-vascular nociception in cat. *Exp Neurol* 1998; 150(1): 45-51.
<http://dx.doi.org/10.1006/exnr.1997.6749> PMID: 9514827
- [125] Muñoz-Islas E, Gupta S, Jiménez-Mena LR, *et al.* Donitriptan, but not sumatriptan, inhibits capsaicin-induced canine external carotid vasodilatation via 5-HT1B rather than 5-HT1D receptors. *Br J Pharmacol* 2006; 149(1): 82-91.
<http://dx.doi.org/10.1038/sj.bjp.0706839> PMID: 16880765
- [126] Muñoz-Islas E, Lozano-Cuenca J, González-Hernández A, *et al.* Spinal sumatriptan inhibits capsaicin-induced canine external carotid vasodilatation via 5-HT1B rather than 5-HT1D receptors. *Eur J Pharmacol* 2009; 615(1-3): 133-8.
<http://dx.doi.org/10.1016/j.ejphar.2009.04.070> PMID: 19460365
- [127] Hougaard A, Amin FM, Christensen CE, *et al.* Increased brainstem perfusion, but no blood-brain barrier disruption, during attacks of migraine with aura. *Brain* 2017; 140(6): 1633-42.
<http://dx.doi.org/10.1093/brain/awx089> PMID: 28430860
- [128] Amin FM, Hougaard A, Cramer SP, *et al.* Intact blood-brain barrier during spontaneous attacks of migraine without aura: a 3T DCE-MRI study. *Eur J Neurol* 2017; 24(9): 1116-24.
<http://dx.doi.org/10.1111/ene.13341> PMID: 28727225
- [129] Schankin CJ, Maniyar FH, Seo Y, *et al.* Ictal lack of binding to brain parenchyma suggests integrity of the blood-brain barrier for 11C-dihydroergotamine during glyceryl trinitrate-induced migraine. *Brain* 2016; 139(Pt 7): 1994-2001.
<http://dx.doi.org/10.1093/brain/aww096> PMID: 27234268
- [130] McCulloch J, Uddman R, Kingman TA, Edvinsson L. Calcitonin gene-related peptide: functional role in cerebrovascular regulation. *Proc Natl Acad Sci USA* 1986; 83(15): 5731-5.
<http://dx.doi.org/10.1073/pnas.83.15.5731> PMID: 3488550
- [131] Nosedá R, Schain AJ, Melo-Carrillo A, *et al.* Fluorescently-labeled fremanezumab is distributed to sensory and autonomic ganglia and the dura but not to the brain of rats with uncompromised blood brain barrier. *Cephalalgia* 2020; 40(3): 229-40.
<http://dx.doi.org/10.1177/0333102419896760> PMID: 31856583
- [132] Johnson KW, Morin SM, Wroblewski VJ, Johnson MP. Peripheral and central nervous system distribution of the CGRP neutralizing antibody [¹²⁵I] galcanezumab in male rats. *Cephalalgia* 2019; 39(10): 1241-8.
<http://dx.doi.org/10.1177/0333102419844711> PMID: 31003588
- [133] Edvinsson JCA, Warfvinge K, Krause DN, *et al.* C-fibers may modulate adjacent Aδ-fibers through axon-axon CGRP signaling at nodes of Ranvier in the trigeminal system. *J Headache Pain* 2019; 20(1): 105.
<http://dx.doi.org/10.1186/s10194-019-1055-3> PMID: 31718551
- [134] Melo-Carrillo A, Strassman AM, Nir RR, *et al.* Fremanezumab-A humanized monoclonal anti-CGRP antibody-inhibits thinly myelinated (Aδ) but not unmyelinated (C) meningeal nociceptors. *J Neurosci* 2017; 37(44): 10587-96.
<http://dx.doi.org/10.1523/JNEUROSCI.2211-17.2017> PMID: 28972120
- [135] Melo-Carrillo A, Schain AJ, Stratton J, Strassman AM, Burstein R. Fremanezumab and its isotype slow propagation rate and shorten cortical recovery period but do not prevent occurrence of cortical spreading depression in rats with compromised blood-brain barrier. *Pain* 2020; 161(5): 1037-43.
<http://dx.doi.org/10.1097/j.pain.0000000000001791> PMID: 31895266
- [136] Pascual J. CGRP antibodies: the Holy Grail for migraine prevention? *Lancet Neurol* 2015; 14(11): 1066-7.
[http://dx.doi.org/10.1016/S1474-4422\(15\)00244-6](http://dx.doi.org/10.1016/S1474-4422(15)00244-6) PMID: 26432183
- [137] Girotra P, Singh SK. Multivariate optimization of rizatriptan benzoate-loaded solid lipid nanoparticles for brain targeting and migraine management. *AAPS PharmSciTech* 2017; 18(2): 517-28.
<http://dx.doi.org/10.1208/s12249-016-0532-0> PMID: 27126007
- [138] Hansraj GP, Singh SK, Kumar P. Sumatriptan succinate loaded chitosan solid lipid nanoparticles for enhanced anti-migraine potential. *Int J Biol Macromol* 2015; 81: 467-76.
<http://dx.doi.org/10.1016/j.ijbiomac.2015.08.035> PMID: 26299709
- [139] Girotra P, Singh SK. A comparative study of orally delivered PBCA and ApoE coupled BSA nanoparticles for brain targeting of sumatriptan succinate in therapeutic management of migraine. *Pharm Res* 2016; 33(7): 1682-95.
<http://dx.doi.org/10.1007/s11095-016-1910-8> PMID: 27003706
- [140] Girotra P, Singh SK, Kumar G. Development of zolmitriptan loaded PLGA/poloxamer nanoparticles for migraine using quality by design approach. *Int J Biol Macromol* 2016; 85: 92-101.
<http://dx.doi.org/10.1016/j.ijbiomac.2015.12.069> PMID: 26724690
- [141] Girotra P, Thakur A, Kumar A, Singh SK. Identification of multi-targeted anti-migraine potential of nystatin and development of its brain targeted chitosan nanoformulation. *Int J Biol Macromol* 2017; 96: 687-96.
<http://dx.doi.org/10.1016/j.ijbiomac.2016.12.065> PMID: 28041915
- [142] Kassem AA. Formulation approaches of triptans for management of migraine. *Curr Drug Deliv* 2016; 13(6): 882-98.
<http://dx.doi.org/10.2174/1566201813666160425112600> PMID: 27109335
- [143] Warfvinge K, Edvinsson L. Distribution of CGRP and CGRP receptor components in the rat brain. *Cephalalgia* 2019; 39(3): 342-53.

- <http://dx.doi.org/10.1177/0333102417728873> PMID: 28856910
- [144] Rubio-Beltrán E, Correnti E, Deen M, *et al.* MaassenVanDenBrink A, Edvinsson L. PACAP38 and PAC1 receptor blockade: a new target for headache? *J Head Pain* 2018; 19: 64.
<http://dx.doi.org/10.1186/s10194-018-0893-8>
- [145] Karatas H, Yemisci M, Eren-Kocak E, Dalkara T. Brain Peptides for the Treatment of Neuropsychiatric Disorders. *Curr Pharm Des* 2018; 24(33): 3905-17.
<http://dx.doi.org/10.2174/1381612824666181112112309> PMID: 30417776
- [146] Alder BioPharmaceuticals®. Announces First-in-Human Dosing in Phase 1 ALD1910 Study for Preventive Treatment of Migraine <https://investor.alderbio.com/news-releases/news-release-details/alder-biopharmaceuticals-announces-first-human-dosing-phase-1>
- [147] Reglodi D, Vaczy A, Rubio-Beltran E, MaassenVanDenBrink A. Protective effects of PACAP in ischemia. *J Head Pain* 2018; 19: 19.
<http://dx.doi.org/10.1186/s10194-018-0845-3>
- [148] Tzabazis A, Kori S, Mechanic J, *et al.* Oxytocin and migraine headache. *Headache* 2017; 57(Suppl. 2): 64-75.
<http://dx.doi.org/10.1111/head.13082> PMID: 28485846
- [149] García-Boll E, Martínez-Lorenzana G, Condés-Lara M, González-Hernández A. Inhibition of nociceptive dural input to the trigemino-cervical complex through oxytocinergic transmission. *Exp Neurol* 2020; 323113079
<http://dx.doi.org/10.1016/j.expneurol.2019.113079> PMID: 31678349
- [150] Strother LC, Srikiatkachorn A, Suprongsinchai W. Targeted orexin and hypothalamic neuropeptides for migraine. *Neurotherapeutics* 2018; 15(2): 377-90.
<http://dx.doi.org/10.1007/s13311-017-0602-3> PMID: 29442286
- [151] González-Hernández A, Condés-Lara M. The multitarget drug approach in migraine treatment: the new challenge to conquer. *Headache* 2014; 54(1): 197-9.
<http://dx.doi.org/10.1111/head.12237> PMID: 24400769
- [152] Goadsby PJ. Primary headache disorders: Five new things. *Neurol Clin Pract* 2019; 9(3): 233-40.
<http://dx.doi.org/10.1212/CPJ.0000000000000654> PMID: 31341711
- [153] Edvinsson L. The CGRP pathway in migraine as a viable target for therapies. *Headache* 2018; 58(Suppl. 1): 33-47.
<http://dx.doi.org/10.1111/head.13305> PMID: 29697153
- [154] Shi L, Lehto SG, Zhu DX, *et al.* Pharmacologic Characterization of AMG 334, a Potent and Selective Human Monoclonal Antibody against the Calcitonin Gene-Related Peptide Receptor. *J Pharmacol Exp Ther* 2016; 356(1): 223-31.
<http://dx.doi.org/10.1124/jpet.115.227793> PMID: 26559125
- [155] Benschop RJ, Gehlert DR, Merchant KM, Shanafelt AB. Treatment of migraine with anti-CGRP antibodies Patent number: WO2007076336 A1, 2007. Assignee: Eli Lilly Co., Benschop RJ, Gehlert DR, Merchant KM, Shanafelt AB
- [156] Zeller J, Poulsen KT, Abdiche YN, Pons J, Collier SJ, Rosenthal A. Antagonist antibodies directed against calcitonin gene-related peptide and methods using same Patent number: WO2007054809, 2007. Assignee: Rinat Neuroscience Corp